

Answers

1.6	2.B	3.A	4.C	5.140	6.C	7.15	8.11
9.C	10.5	11.4	12.4	13.0	14.D	15.12	16.A
17.C	18.B	19.D	20.C	21.45	22.50	23.35	24.D
25.9	26.C	27.A	28.A	29.B	30.C	31.A	32.B
33.C	34.A	35.D	36.C	37.D	38.D	39.A	40.3
41.3	42.D	43.2	44.B	45.D	46.C	47.C	48.D
49.A	50.B	51.B	52.D	53.B	54.C	55.D	56.C
57.A	58.B	59.B	60.C	61.A	62.D	63.4	64.A
65.B	66.84	67.D	68.A	69.A	70.40	71.2	72.26
73.C	74.4	75.A	76.6	77.4	78.B	79.5	80.B

Explanations

1.6

Consider Statement 2: Anjali took Ride-1 at 11 am after waiting for 30 minutes for Chitra to complete it. It was the only ride where Anjali waited.

This implies that Chitra took Ride 1 at 10 am. Now we also know that she spent Rs 50 and that she left at 11 am. Now, since she did one ride costing Rs 20 at 10, she must have taken Ride-3 at 9 am.

So we get the following table for Chitra.

	Ride 3	Ride 1
Time	9 am- 10 am	10 am- 11 am
Cost	Rs 30	Rs 20

Now we know that Chitra and Anjali spent Rs 50 before 12:15 pm. It is not possible for Anjali to go on Ride-3 at 10 am as we know that she was waiting for 30 minutes before taking Ride-1 (She was waiting from 10:30 am).

Now, since we know that Ride-1 was the only ride for which she waited, we can say that she took Ride-1 at 11 am and started Ride-3 at 12 am

So we get the following table for Anjali.

	Ride-1	Ride-3
Time	11 am- 12 pm	12 pm - 1pm
Cost	20	30

Now, we know that Bipasha started her first ride at 11:30 am. We also know that they all spent Rs 50 before 12:15 pm.

Therefore, the first ride Bipasha takes will be Ride-2, costing Rs 50.

So we get the following table for Bipasha.

	Ride-2	
Time	11:30 am- 12:30am	
Cost	50	

We know that Ride 3 stops at 1 pm. So the last ride taken by Anjali will either be Ride-2 or Ride-4. Now, considering Statement 4, we know that the last ride taken by Anjali and Bipasha was same and that Bipasha rode it after Anjali. So their last ride can't be 2.

So the last ride of both Bipasha and Anjali will be 4.

Now if we assume that immediately after ending Ride-3, Anjali goes to Ride-4, then the last ride of Bipasha will be Ride-4 from 2 pm - 3 pm. But we know that Bipasha rode 3 rides. So this case is not possible.

Since Anjali didn't have any break or waiting time, the only ride she can ride at 1 pm will be Ride 2 and then she will go on Ride-4 from 2 pm to 3 pm.

So we get the following table for Anjali:

	Ride-1	Ride-3	Ride-2	Ride-4
Time	11 am- 12 pm	12 pm -1 pm	1 pm- 2 pm	2 pm- 3pm
Cost	20	30	50	40

Now we know that the last ride that Bipasha took was Ride-4 and that she had a gap of 1.5 hrs before it. This is only possible when she takes one ride between Ride-2 and Ride-4. Since Ride-3 is closed at 1 pm, she can only take Ride 1. So we get the following table for her.

	Ride-2	Ride-1	Break	Waiting time	Ride-4
Time	11:30 am-12:30 pm	12:30 pm- 1:30 pm	1:30 pm- 2:30 pm	2:30 pm to 3:00pm	3 pm - 4 pm
Cost	50	20			40

Anjali took 4 rides, and Chitra took 2 rides. Therefore the correct answer is 6

 VIDEO SOLUTION

2. B

Consider Statement 2: Anjali took Ride-1 at 11 am after waiting for 30 minutes for Chitra to complete it. It was the only ride where Anjali waited.

This implies that Chitra took Ride 1 at 10 am. Now we also know that she spent Rs 50 and that she left at 11 am. Now, since she did one ride costing Rs 20 at 10, she must have taken Ride-3 at 9 am.

So we get the following table for Chitra.

	Ride 3	Ride 1
Time	9 am- 10 am	10 am- 11 am
Cost	Rs 30	Rs 20

Now we know that Chitra and Anjali spent Rs 50 before 12:15 pm. It is not possible for Anjali to go on Ride-3 at 10 am as we know that she was waiting for 30 minutes before taking Ride-1 (She was waiting from 10:30 am).

Now, since we know that Ride-1 was the only ride for which she waited, we can say that she took Ride-1 at 11 am and started Ride-3 at 12 am

So we get the following table for Anjali.

	Ride-1	Ride-3
Time	11 am- 12 pm	12 pm - 1pm
Cost	20	30

Now, we know that Bipasha started her first ride at 11:30 am. We also know that they all spent Rs 50 before 12:15 pm.

Therefore, the first ride Bipasha takes will be Ride-2, costing Rs 50.

So we get the following table for Bipasha.

	Ride-2	
Time	11:30 am- 12:30am	
Cost	50	

We know that Ride 3 stops at 1 pm. So the last ride taken by Anjali will either be Ride-2 or Ride-4. Now, considering Statement 4, we know that the last ride taken by Anjali and Bipasha was same and that Bipasha rode it after Anjali. So their last ride can't be 2.

So the last ride of both Bipasha and Anjali will be 4.

Now if we assume that immediately after ending Ride-3, Anjali goes to Ride-4, then the last ride of Bipasha will be Ride-4 from 2 pm - 3 pm. But we know that Bipasha rode 3 rides. So this case is not possible.

Since Anjali didn't have any break or waiting time, the only ride she can ride at 1 pm will be Ride 2 and then she will go on Ride-4 from 2 pm to 3 pm.

So we get the following table for Anjali:

	Ride-1	Ride-3	Ride-2	Ride-4
Time	11 am- 12 pm	12 pm -1 pm	1 pm- 2 pm	2 pm- 3pm
Cost	20	30	50	40

Now we know that the last ride that Bipasha took was Ride-4 and that she had a gap of 1.5 hrs before it. This is only possible when she takes one ride between Ride-2 and Ride-4. Since Ride-3 is closed at 1 pm, she can only take Ride 1. So we get the following table for her.

	Ride-2	Ride-1	Break	Waiting time	Ride-4
Time	11:30 am-12:30 pm	12:30 pm- 1:30 pm	1:30 pm- 2:30 pm	2:30 pm to 3:00pm	3 pm - 4 pm
Cost	50	20			40

Anjali completed a total of 4 rides, 3 of which were completed at 2. Therefore the answer is Option B: Ride-1, Ride 3, and Ride -2

[VIDEO SOLUTION](#)

3. A

Consider Statement 2: Anjali took Ride-1 at 11 am after waiting for 30 minutes for Chitra to complete it. It was the only ride where Anjali waited.

This implies that Chitra took Ride 1 at 10 am. Now we also know that she spent Rs 50 and that she left at 11 am. Now, since she did one ride costing Rs 20 at 10, she must have taken Ride-3 at 9 am.

So we get the following table for Chitra.

	Ride 3	Ride 1
Time	9 am- 10 am	10 am- 11 am
Cost	Rs 30	Rs 20

Now we know that Chitra and Anjali spent Rs 50 before 12:15 pm. It is not possible for Anjali to go on Ride-3 at 10 am as we know that she was waiting for 30 minutes before taking Ride-1 (She was waiting from 10:30 am).

Now, since we know that Ride-1 was the only ride for which she waited, we can say that she took Ride-1 at 11 am and started Ride-3 at 12 am

So we get the following table for Anjali.

	Ride-1	Ride-3
Time	11 am- 12 pm	12 pm - 1pm
Cost	20	30

Now, we know that Bipasha started her first ride at 11:30 am. We also know that they all spent Rs 50 before 12:15 pm.

Therefore, the first ride Bipasha takes will be Ride-2, costing Rs 50.

So we get the following table for Bipasha.

	Ride-2	
Time	11:30 am- 12:30am	
Cost	50	

We know that Ride 3 stops at 1 pm. So the last ride taken by Anjali will either be Ride-2 or Ride-4. Now, considering Statement 4, we know that the last ride taken by Anjali and Bipasha was same and that Bipasha rode it after Anjali. So their last ride can't be 2.

So the last ride of both Bipasha and Anjali will be 4.

Now if we assume that immediately after ending Ride-3, Anjali goes to Ride-4, then the last ride of Bipasha will be Ride-4 from 2 pm - 3 pm. But we know that Bipasha rode 3 rides. So this case is not possible.

Since Anjali didn't have any break or waiting time, the only ride she can ride at 1 pm will be Ride 2 and then she will go on Ride-4 from 2 pm to 3 pm.

So we get the following table for Anjali:

	Ride-1	Ride-3	Ride-2	Ride-4
Time	11 am- 12 pm	12 pm -1 pm	1 pm- 2 pm	2 pm- 3pm
Cost	20	30	50	40

Now we know that the last ride that Bipasha took was Ride-4 and that she had a gap of 1.5 hrs before it. This is only possible when she takes one ride between Ride-2 and Ride-4. Since Ride-3 is closed at 1 pm, she can only take Ride 1. So we get the following table for her.

	Ride-2	Ride-1	Break	Waiting time	Ride-4
Time	11:30 am-12:30 pm	12:30 pm- 1:30 pm	1:30 pm- 2:30 pm	2:30 pm to 3:00pm	3 pm - 4 pm
Cost	50	20			40

Only Ride-1 was taken by all the visitors. Therefore the correct answer is Option A: Ride-1

[VIDEO SOLUTION](#)

4. C

Consider Statement 2: Anjali took Ride-1 at 11 am after waiting for 30 minutes for Chitra to complete it. It was the only ride where Anjali waited.

This implies that Chitra took Ride 1 at 10 am. Now we also know that she spent Rs 50 and that she left at 11 am. Now, since she did one ride costing Rs 20 at 10, she must have taken Ride-3 at 9 am.

So we get the following table for Chitra.

	Ride 3	Ride 1
Time	9 am- 10 am	10 am- 11 am
Cost	Rs 30	Rs 20

Now we know that Chitra and Anjali spent Rs 50 before 12:15 pm. It is not possible for Anjali to go on Ride-3 at 10 am as we know that she was waiting for 30 minutes before taking Ride-1 (She was waiting from 10:30 am).

Now, since we know that Ride-1 was the only ride for which she waited, we can say that she took Ride-1 at 11 am and started Ride-3 at 12 am

So we get the following table for Anjali.

	Ride-1	Ride-3
Time	11 am- 12 pm	12 pm - 1pm
Cost	20	30

Now, we know that Bipasha started her first ride at 11:30 am. We also know that they all spent Rs 50 before 12:15 pm.

Therefore, the first ride Bipasha takes will be Ride-2, costing Rs 50.

So we get the following table for Bipasha.

	Ride-2	
Time	11:30 am- 12:30am	
Cost	50	

We know that Ride 3 stops at 1 pm. So the last ride taken by Anjali will either be Ride-2 or Ride-4. Now, considering Statement 4, we know that the last ride taken by Anjali and Bipasha was same and that Bipasha rode it after Anjali. So their last ride can't be 2.

So the last ride of both Bipasha and Anjali will be 4.

Now if we assume that immediately after ending Ride-3, Anjali goes to Ride-4, then the last ride of Bipasha will be Ride-4 from 2 pm - 3 pm. But we know that Bipasha rode 3 rides. So this case is not possible.

Since Anjali didn't have any break or waiting time, the only ride she can ride at 1 pm will be Ride 2 and then she will go on Ride-4 from 2 pm to 3 pm.

So we get the following table for Anjali:

	Ride-1	Ride-3	Ride-2	Ride-4
Time	11 am- 12 pm	12 pm -1 pm	1 pm- 2 pm	2 pm- 3pm
Cost	20	30	50	40

Now we know that the last ride that Bipasha took was Ride-4 and that she had a gap of 1.5 hrs before it. This is only possible when she takes one ride between Ride-2 and Ride-4. Since Ride-3 is closed at 1 pm, she can only take Ride 1. So we get the following table for her.

	Ride-2	Ride-1	Break	Waiting time	Ride-4
Time	11:30 am-12:30 pm	12:30 pm- 1:30 pm	1:30 pm- 2:30 pm	2:30 pm to 3:00pm	3 pm - 4 pm
Cost	50	20			40

As we can see from the table for Bipasha, she spent a total of $50+20+40=110$

 VIDEO SOLUTION

CAT Percentile Predictor



6. C

Input of the scanned sequence of five fingers is allowed to vary from the original sequence by one place for any of the fingers. This can be achieved only when two consecutive fingers are interchanged. Let the original sequence be TIMRL

Case 1: Only a set of two consecutive numbers are interchanged.

They can be selected in $5-1 = 4$ ways

Case 2: Two sets of two consecutive numbers are interchanged.

(i) TI are interchanged, $\Rightarrow$ (MR, RL) $\Rightarrow$ 2 ways

(ii) IM are interchanged $\Rightarrow$ (RL) $\Rightarrow$ 1 way

Total no of ways possible = $4 + 2 + 1 = 7$

Including the original sequence, we get the total number of allowed combinations as 8

7. 15

There can be two scans out of place.

TIMTRL is the original sequence.

If T is interchanged: There will be four ways: ITMTRL, MITTRL, RIMTTL, LIMTRT

If I is interchanged: There will be four ways

If M is interchanged: There will be three ways

If T is interchanged: There will be two ways

If R is interchanged: There will be one way

Total 14.

Another sequence allowed is original, So total 15 ways.

8. 11

Let the original sequence be TIMRL

Two fingers can be out of place. This can be done if and only if two fingers interchange their position. These two can be selected in ${}^5C_2 = 10$ ways. In addition to these, the original sequence will also be accepted. Hence the total number of acceptable sequences = $10 + 1 = 11$

9. C

1. If original sequence is given.

2. If either of LR, RL, LT, TI, IM is interchanged $\Rightarrow$ 5 ways.

3. If LR and LT and IM interchanged. The sequence will look like: RLTLMI

4. If LR and LT are interchanged.

5. If LR and TI are interchanged.

6. If LR and IM are interchanged.
7. If RL and TI are interchanged.
8. If RL and IM are interchanged.
9. If LT and IM are interchanged.

Total 13 ways possible.

 VIDEO SOLUTION

10.5

Denoting Asha, Bunty, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novice after Q1, and B cannot go back to Elite after Q2 and come back to Novice after Q3 because B came to Novice in Q1, and A came to Novice in Q3, so C has to come to Novice after Q3 to be present in Novice at the start of Q4 as per the table. Therefore, we can conclude that C came to Novice after Q3, and B stayed in Novice in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novice after Q2. So, we can conclude that D stayed in Novice during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novice after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novice in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novice after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novice from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novice and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novice, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Bunty's score after quarter 3 is 5.

Hence, the correct answer is 5.



11.4

Denoting Asha, Bunty, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novice after Q1, and B cannot go back to Elite after Q2 and come back to Novice after Q3 because B came to Novice in Q1, and A came to Novice in Q3, so C has to come to Novice after Q3 to be present in Novice at the start of Q4 as per the table. Therefore, we can conclude that C came to Novice after Q3, and B stayed in Novice in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novice after Q2. So, we can conclude that D stayed in Novice during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novice after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novice in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novice after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novice from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novice and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novice, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Ekalavya's score after quarter 2 is 4.

Hence, the correct answer is 4.

12.4

Denoting Asha, Bunt, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novice after Q1, and B cannot go back to Elite after Q2 and come back to Novice after Q3 because B came to Novice in Q1, and A came to Novice in Q3, so C has to come to Novice after Q3 to be present in Novice at the start of Q4 as per the table. Therefore, we can conclude that C came to Novice after Q3, and B stayed in Novice in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novice after Q2. So, we can conclude that D stayed in Novice during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novice after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novice in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novice after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novice from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novice and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novice, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

The cumulative score of 4 employees can be determined with certainty at the end of quarter 3.

Hence, the correct answer is 4.

13.0

Denoting Asha, Bunt, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novice after Q1, and B cannot go back to Elite after Q2 and come back to Novice after Q3 because B came to Novice in Q1, and A came to Novice in Q3, so C has to come to Novice after Q3 to be present in Novice at the start of Q4 as per the table. Therefore, we can conclude that C came to Novice after Q3, and B stayed in Novice in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novice after Q2. So, we can conclude that D stayed in Novice during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novice after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novice in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novice after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novice from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novice and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novice, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

As we can see in the table, no employee has changed their group more than once.

Hence, the correct answer is 0.

14. D

Denoting Asha, Bunt, Chintu, Dolly, Eklavya, and Falguni as A, B, C, D, E and F for easy usage. The values in the brackets are the ratings obtained in that quarter, and the values outside are the cumulative ratings after that Quarter.

Putting all the given information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A			D
B			E
C			F

Novie(Lalu)			
Q1	Q2	Q3	Q4
D	B(1)	A(1)	A
E		D(2)	B
F			C

B came from Elite to Novice after Q1, and B cannot go back to Elite after Q2 and come back to Novice after Q3 because B came to Novice in Q1, and A came to Novice in Q3, so C has to come to Novice after Q3 to be present in Novice at the start of Q4 as per the table. Therefore, we can conclude that C came to Novice after Q3, and B stayed in Novice in Q3 as well, receiving a rating of 3 in Q3. Similarly, D cannot go to Elite after Q1 and come back after Q2 because we already know that A is coming to Novice after Q2. So, we can conclude that D stayed in Novice during Q2 as well. We are also given that A and F received the same rating during all the quarters, and we know that the rating of D is 2 in Q3, so we can conclude its rating to be 2 in all the quarters. For B to come to Novice after Q1, it must receive a rating of 1 in Q1.

We are also given that F has the same rating across all the quarters, and it has to be either 1 or 3, as D already has a rating of 2 in Q1 in that group. If the rating of F is 1, then it will stay in Novice in quarter 2 and also will get a rating of 1 in Q2, which makes the cumulative score 2. But we know that both E and F are going to elite after quarters 1 and 2 in some order, and if the cumulative rating of F is 2 at the end of Q2, then it is not possible for it to go to elite after Q2, as there are members with ratings higher than 2 in Novice after Q2.

Therefore, for all conditions to be satisfied, the rating of F must be 3 in all quarters, and F advances to Elite after Q1 and remains there. Also, we can conclude that E goes to Elite after Q2. Putting all the information in the table, we get,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A	A	C	D(2) - 8
B(1) - 1	C	E	E
C	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

Now, if we look at the Elite table, we know that A and C received 2 and 3 in some order in Q1 and 1 and 2 in Q2 in some order.

We know that A goes to Novice from Elite after Q2, so the cumulative rating of A must either be less than C after Q2 or equal to C, and the rating of A in Q2 is less than C. These are the two possibilities.

If A received a rating of 3 in Q1, then C gets 2 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 4 after 2 quarters which is same as the cumulative rating of C after 2 quarters and because C gets a rating of 2 and A gets a rating of 1 in Q2, A goes to Novice and C stays in elite even with the same cumulative rating.

If A received a rating of 2 in Q1, then C gets 3 in Q1. For A to get to Novice after Q2, it must obtain a rating of 1 in Q2, so that it will have a cumulative rating of 3 after 2 quarters, which is less than the cumulative rating of C after 2 quarters. If it gets a rating of 2 in Q2, then its cumulative becomes 4, which would be the same as C in that case, and because the rating in Q2 would be less for C in that case, it gets demoted to Novice, which is not the case. So, in the case of A getting 2 in Q1, it must get 1 in Q2.

After Q2, E and C will either have the same cumulative score or C will have 1 point more than E's cumulative score. For E to remain in Elite even after Q3, it must get a rating of 2 points in Q3, and C must get a rating of 1 point.

Putting all the calculated values, we get the final table as,

Elite(Kuku)			
Q1	Q2	Q3	Q4
A(3/2) - 3/2	A(1) - 4/3	C(1) - 5/6	D(2) - 8
B(1) - 1	C(2) - 4/5	E(2) - 6	E(1) - 7
C(2/3) - 2/3	F(3) - 6	F(3) - 9	F(3) - 12

Novie(Lalu)			
Q1	Q2	Q3	Q4
D(2) - 2	B(1) - 2	A(1) - (5/4)	A
E(1) - 1	D(2) - 4	D(2) - 6	B
F(3) - 3	E(3) - 4	B(3) - 5	C

I. Asha received a rating of 2 in Quarter 1 - This is not necessarily true, as it can also be 3 in Quarter 1.

II. Asha received a rating of 1 in Quarter 2. - This is definitely true, as Asha received a rating of 1 in Quarter 1.

Only II is definitely correct.

Hence, the correct answer is option D.

 VIEW SOLUTION

15.12

We have the following table :

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

o10 is given to Elise and o9 is given to Bharat .

Now as Elise values his own bundle at an even number so the only two objects which can be given to Elise is o1 or o5 or o7.

Case 1 :

o1 is given to Elise

Now the total valuation of Elise = 12

Valuation of Disha is an odd number

So we can say Amar , Bharat and Charles values their bundles at 16 .

So for Bharat the valuation to be 16, o7 will be given to him

so we get

Bharat - o9 and o7 and Elise -o10 and o1

For charles to have valuation 16

the only way = 8+8

so we can say o8 is given to charles along with either o2 or o3 .(o1 and o8 cannot be together)

Now for Amart to have a valuation of 16

the only way possible = 9+7

Now so we can say

Amar will receive either o2 or o3 and o5 .

Now we are left with 04 and o6

So if Disha receives o4 and o6

The valuation of Disha will be 5+3 =8 which is not an odd number

so this case is discarded.

Case 2 Elise receives o5 or o7 .

Now Valuation of Elise = 16 .

And Elise receives o10 and o5/o7.

Bharat received o9 and we know the evaluation of Bharat is an even number and the minimum even number possible for valuation of Bharat is 16 and no one can have evaluation more than 16 so Bharat received o7 .

So Elise received o5 .

So we have

Bharat - o9 ,o7

Elise -o10,o5.

Now as we know o1 ,o2 and o3 are given to three different persons so they are Amar, Charles and Disha .

Now As per Amar

he values Bharat at 17 so he envy him

So Amar will value his bundle less than 16

So the only possibility for Amar to value his bundle less than 16 = 12 =9+3.

Now we can say Charu will have 16 as his own valuation so he will get 8+8 .

Now o8 will be given to Charu, and he cannot have o1 , also he cannot have o2 because if he has o2 he will value Bharat's bundle as 17 and will envy him which is not possible so Charu will have o3,o8

Now Amar will have o2 and Disha will have o1.

Now Amar will not have o4 because in that case Charles will envy Amar and is not possible so we can say Amar will have o6 and Disha will have o4.

So we have the following :

Amar - o2,o6

Bharat -o9,o7

Charu -o3,o8

Disha o1,o4

Elise -o10,o5

Amar's own valuation = $9+3=12$

 VIDEO SOLUTION



16. A

We have the following table :

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

o10 is given to Elise and o9 is given to Bharat .

Now as Elise values his own bundle at an even number so the only two objects which can be given to Elise is o1 or o5 or o7.

Case 1 :

o1 is given to Elise

Now the total valuation of Elise = 12

Valuation of Disha is an odd number

So we can say Amar , Bharat and Charles values their bundles at 16 .

So for Bharat the valuation to be 16, o7 will be given to him

so we get

Bharat - o9 and o7 and Elise -o10 and o1

For charles to have valuation 16

the only way = $8+8$

so we can say o8 is given to charles along with either o2 or o3 .(o1 and o8 cannot be together)

Now for Amart to have a valuation of 16

the only way possible = $9+7$

Now so we can say

Amar will receive either o2 or o3 and o5 .

Now we are left with 04 and o6

So if Disha receives o4 and o6

The valuation of Disha will be $5+3=8$ which is not an odd number

so this case is discarded.

Case 2 Elise receives o5 or o7 .

Now Valuation of Elise = 16 .

And Elise receives o10 and o5/o7.

Bharat received o9 and we know the evaluation of Bharat is an even number and the minimum even number possible for valuation of Bharat is 16 and no one can have evaluation more than 16 so Bharat received o7 .

So Elise received o5 .

So we have

Bharat - o9 ,o7

Elise -o10,o5.

Now as we know o1 ,o2 and o3 are given to three different persons so they are Amar, Charles and Disha .

Now As per Amar

he values Bharat at 17 so he envy him

So Amar will value his bundle less than 16

So the only possibility for Amar to value his bundle less than 16 = 12 =9+3.

Now we can say Charu will have 16 as his own valuation so he will get 8+8 .

Now o8 will be given to Charu, and he cannot have o1 , also he cannot have o2 because if he has o2 he will value Bharat's bundle as 17 and will envy him which is not possible so Charu will have o3,o8

Now Amar will have o2 and Disha will have o1.

Now Amar will not have o4 because in that case Charles will envy Amar and is not possible so we can say Amar will have o6 and Disha will have o4.

So we have the following :

Amar - o2,o6

Bharat -o9,o7

Charu -o3,o8

Disha o1,o4

Elise -o10,o5

So o1 is given to Disha

 VIDEO SOLUTION

17. C

We have the following table :

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

o10 is given to Elise and o9 is given to Bharat .

Now as Elise values his own bundle at an even number so the only two objects which can be given to Elise is o1 or o5 or o7.

Case 1 :

o1 is given to Elise

Now the total valuation of Elise = 12

Valuation of Disha is an odd number

So we can say Amar , Bharat and Charles values their bundles at 16 .

So for Bharat the valuation to be 16, o7 will be given to him

so we get

Bharat - o9 and o7 and Elise -o10 and o1

For charles to have valuation 16

the only way = 8+8

so we can say o8 is given to charles along with either o2 or o3 .(o1 and o8 cannot be together)

Now for Amart to have a valuation of 16

the only way possible = 9+7

Now so we can say

Amar will receive either o2 or o3 and o5 .

Now we are left with O4 and o6

So if Disha receives o4 and o6

The valuation of Disha will be $5+3=8$ which is not an odd number
so this case is discarded.

Case 2 Elise receives o5 or o7 .

Now Valuation of Elise = 16 .

And Elise receives o10 and o5/o7.

Bharat received o9 and we know the evaluation of Bharat is an even number and the minimum even number possible for valuation of Bharat is 16 and no one can have evaluation more than 16 so Bharat received o7 .

So Elise received o5 .

So we have

Bharat - o9 ,o7

Elise -o10,o5.

Now as we know o1 ,o2 and o3 are given to three different persons so they are Amar, Charles and Disha .

Now As per Amar

he values Bharat at 17 so he envy him

So Amar will value his bundle less than 16

So the only possibility for Amar to value his bundle less than 16 = $12=9+3$.

Now we can say Charu will have 16 as his own valuation so he will get $8+8$.

Now o8 will be given to Charu, and he cannot have o1 , also he cannot have o2 because if he has o2 he will value Bharat's bundle as 17 and will envy him which is not possible so Charu will have o3,o8

Now Amar will have o2 and Disha will have o1.

Now Amar will not have o4 because in that case Charles will envy Amar and is not possible so we can say Amar will have o6 and Disha will have o4.

So we have the following :

Amar - o2,o6

Bharat -o9,o7

Charu -o3,o8

Disha o1,o4

Elise -o10,o5

So Amar envies someone else

 VIDEO SOLUTION

18. B

We have the following table :

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

o10 is given to Elise and o9 is given to Bharat .

Now as Elise values his own bundle at an even number so the only two objects which can be given to Elise is o1 or o5 or o7.

Case 1 :

o1 is given to Elise

Now the total valuation of Elise = 12

Valuation of Disha is an odd number

So we can say Amar , Bharat and Charles values their bundles at 16 .

So for Bharat the valuation to be 16, o7 will be given to him

so we get

Bharat - o9 and o7 and Elise -o10 and o1

For Charles to have valuation 16

the only way = $8+8$

so we can say o8 is given to Charles along with either o2 or o3. (o1 and o8 cannot be together)

Now for Amar to have a valuation of 16

the only way possible = $9+7$

Now so we can say

Amar will receive either o2 or o3 and o5.

Now we are left with o4 and o6

So if Disha receives o4 and o6

The valuation of Disha will be $5+3=8$ which is not an odd number
so this case is discarded.

Case 2 Elise receives o5 or o7.

Now Valuation of Elise = 16.

And Elise receives o10 and o5/o7.

Bharat received o9 and we know the evaluation of Bharat is an even number and the minimum even number possible for valuation of Bharat is 16 and no one can have evaluation more than 16 so Bharat received o7.

So Elise received o5.

So we have

Bharat - o9, o7

Elise - o10, o5.

Now as we know o1, o2 and o3 are given to three different persons so they are Amar, Charles and Disha.

Now As per Amar

he values Bharat at 17 so he envies him

So Amar will value his bundle less than 16

So the only possibility for Amar to value his bundle less than 16 = $12=9+3$.

Now we can say Charu will have 16 as his own valuation so he will get $8+8$.

Now o8 will be given to Charu, and he cannot have o1, also he cannot have o2 because if he has o2 he will value Bharat's bundle as 17 and will envy him which is not possible so Charu will have o3, o8

Now Amar will have o2 and Disha will have o1.

Now Amar will not have o4 because in that case Charles will envy Amar and is not possible so we can say Amar will have o6 and Disha will have o4.

So we have the following :

Amar - o2, o6

Bharat - o9, o7

Charu - o3, o8

Disha o1, o4

Elise - o10, o5

o5 is given to Elise

 VIDEO SOLUTION

19. D

We have the following table :

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

o10 is given to Elise and o9 is given to Bharat .

Now as Elise values his own bundle at an even number so the only two objects which can be given to Elise is o1 or o5 or o7.

Case 1 :

o1 is given to Elise

Now the total valuation of Elise = 12

Valuation of Disha is an odd number

So we can say Amar , Bharat and Charles values their bundles at 16 .

So for Bharat the valuation to be 16, o7 will be given to him

so we get

Bharat - o9 and o7 and Elise -o10 and o1

For charles to have valuation 16

the only way = $8+8$

so we can say o8 is given to charles along with either o2 or o3 .(o1 and o8 cannot be together)

Now for Amart to have a valuation of 16

the only way possible = $9+7$

Now so we can say

Amar will receive either o2 or o3 and o5 .

Now we are left with 04 and o6

So if Disha receives o4 and o6

The valuation of Disha will be $5+3=8$ which is not an odd number

so this case is discarded.

Case 2 Elise receives o5 or o7 .

Now Valuation of Elise = 16 .

And Elise receives o10 and o5/o7.

Bharat received o9 and we know the evaluation of Bharat is an even number and the minimum even number possible for valuation of Bharat is 16 and no one can have evaluation more than 16 so Bharat received o7 .

So Elise received o5 .

So we have

Bharat - o9 ,o7

Elise -o10,o5.

Now as we know o1 ,o2 and o3 are given to three different persons so they are Amar, Charles and Disha .

Now As per Amar

he values Bharat at 17 so he envy him

So Amar will value his bundle less than 16

So the only possibility for Amar to value his bundle less than 16 = $12=9+3$.

Now we can say Charu will have 16 as his own valuation so he will get $8+8$.

Now o8 will be given to Charu, and he cannot have o1 , also he cannot have o2 because if he has o2 he will value Bharat's bundle as 17 and will envy him which is not possible so Charu will have o3,o8

Now Amar will have o2 and Disha will have o1.

Now Amar will not have o4 because in that case Charles will envy Amar and is not possible so we can say Amar will have o6 and Disha will have o4.

So we have the following :

Amar - o2,o6

Bharat -o9,o7

Charu -o3,o8

Disha o1,o4

Elise -o10,o5

o4 is given to Disha



20.C

We have the following table :

	o1	o2	o3	o4	o5	o6	o7	o8	o9	o10
Amar	4	9	9	3	7	3	8	7	9	5
Barat	5	9	7	5	5	3	6	8	10	8
Charles	8	8	8	3	6	4	5	8	9	6
Disha	8	8	8	5	5	3	6	4	9	8
Elise	6	8	9	5	6	5	6	3	7	10

o10 is given to Elise and o9 is given to Bharat .

Now as Elise values his own bundle at an even number so the only two objects which can be given to Elise is o1 or o5 or o7.

Case 1 :

o1 is given to Elise

Now the total valuation of Elise = 12

Valuation of Disha is an odd number

So we can say Amar , Bharat and Charles values their bundles at 16 .

So for Bharat the valuation to be 16, o7 will be given to him

so we get

Bharat - o9 and o7 and Elise -o10 and o1

For charles to have valuation 16

the only way = 8+8

so we can say o8 is given to charles along with either o2 or o3 .(o1 and o8 cannot be together)

Now for Amart to have a valuation of 16

the only way possible = 9+7

Now so we can say

Amar will receive either o2 or o3 and o5 .

Now we are left with O4 and o6

So if Disha receives o4 and o6

The valuation of Disha will be 5+3 =8 which is not an odd number

so this case is discarded.

Case 2 Elise receives o5 or o7 .

Now Valuation of Elise = 16 .

And Elise receives o10 and o5/o7.

Bharat received o9 and we know the evaluation of Bharat is an even number and the minimum even number possible for valuation of Bharat is 16 and no one can have evaluation more than 16 so Bharat received o7 .

So Elise received o5 .

So we have

Bharat - o9 ,o7

Elise -o10,o5.

Now as we know o1 ,o2 and o3 are given to three different persons so they are Amar, Charles and Disha .

Now As per Amar

he values Bharat at 17 so he envy him

So Amar will value his bundle less than 16

So the only possibility for Amar to value his bundle less than 16 = 12 =9+3.

Now we can say Charu will have 16 as his own valuation so he will get 8+8 .

Now o8 will be given to Charu, and he cannot have o1 , also he cannot have o2 because if he has o2 he will value Bharat's bundle as 17 and will envy him which is not possible so Charu will have o3,o8

Now Amar will have o2 and Disha will have o1.

Now Amar will not have o4 because in that case Charles will envy Amar and is not possible so we can say Amar will have o6 and Disha will have o4.

So we have the following :

Amar - o2,o6

Bharat -o9,o7

Charu -o3,o8

Disha o1,o4

Elise -o10,o5

So o8 is given to Charu.

 VIDEO SOLUTION



21.45

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humbleset. So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humbleset has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humbleset.

We can also conclude that the PMs of Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo are 50, 60, 70, 80, and 90, respectively.

We are given that in the PI calculation, cities have a contribution of 25% and NUR has a contribution of 50%. So, we can calculate the contribution of each city in the calculation of PI by dividing their PI value by 4, and the contribution of NUR can be calculated by dividing the PM value by 2.

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50		12.5
Splutterville	60		15
Quackford	70		17.5
Mumpypore	80		20
Zingaloo	90		22.5
NUR(Whimshire)	20/10	Whimshire	10/5
NUR(Foggia)	10/20	Foggia	5/10
NUR(Humbleset)	40	Humbleset	20

We are given that all the PI values are unique integers, and Humbleset has the highest PI value, while Foggia has the lowest.

For the PI value to be an integer, the cities with decimal contributions must be paired with another city with decimal contributions. Since all the decimals are 0.5, when two cities with 0.5 are added, the sum becomes 1, making the PI an integer.

The cities Splutterville and Mumpypore must be from the same state; both have integral contributions towards the PI, and if they are added to a decimal, then the result will be a decimal, which should not be the case. These two cities have to belong to either the state of Whimshire or Foggia, but their combined contribution towards PI is $15 + 20 = 35$.

We already know that for the state of Humbleset, the PI is highest, and the two contributions towards the PI are 7.5 and 20, totalling 27.5.

There are three possibilities for the other city of Humbleset, which are Noodleton, Quackford and Zingaloo.

CASE 1: Noodleton

If the other city of Humbleset is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

CASE 2: Quackford

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

CASE 3: Zingaloo

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

Then, for Humbleset, the PI is $7.5 + 22.5 + 20 = 50$.

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

If the PI of the other state is 45,

For one of the other 2 states, the PI is $15 + 20 + 10 = 45$

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humbleset, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

66/150

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50	Foggia	12.5
Splutterville	60	Whimshire	15
Quackford	70	Foggia	17.5
Mumpypore	80	Whimshire	20
Zingaloo	90	Humbleset	22.5
NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humbleset)	40	Humbleset	20

We calculated the PI of Whimshire to be 45.

Hence, the correct answer is 45.

[VIEW SOLUTION](#)

22. 50

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humbleset. So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humbleset has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humbleset.

We can also conclude that the PMs of Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo are 50, 60, 70, 80, and 90, respectively.

We are given that in the PI calculation, cities have a contribution of 25% and NUR has a contribution of 50%. So, we can calculate the contribution of each city in the calculation of PI by dividing their PI value by 4, and the contribution of NUR can be calculated by dividing the PM value by 2.

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50		12.5
Splutterville	60		15
Quackford	70		17.5
Mumpypore	80		20
Zingaloo	90		22.5
NUR(Whimshire)	20/10	Whimshire	10/5
NUR(Foggia)	10/20	Foggia	5/10
NUR(Humbleset)	40	Humbleset	20

We are given that all the PI values are unique integers, and Humbleset has the highest PI value, while Foggia has the lowest.

For the PI value to be an integer, the cities with decimal contributions must be paired with another city with decimal contributions. Since all the decimals are 0.5, when two cities with 0.5 are added, the sum becomes 1, making the PI an integer.

The cities Splutterville and Mumpypore must be from the same state; both have integral contributions towards the PI, and if they are added to a decimal, then the result will be a decimal, which should not be the case. These two cities have to belong to either the state of Whimshire or Foggia, but their combined contribution towards PI is $15 + 20 = 35$.

We already know that for the state of Humbleset, the PI is highest, and the two contributions towards the PI are 7.5 and 20, totalling 27.5.

There are three possibilities for the other city of Humbleset, which are Noodleton, Quackford and Zingaloo.

CASE 1: Noodleton

If the other city of Humblest is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humblest.

CASE 2: Quackford

If the other city of Humblest is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humblest, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humblest.

CASE 3: Zingaloo

If the other city of Humblest is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

If the PI of the other state is 40,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

We can see that PI of the other two states is 40, which violates one of the conditions given.

If the PI of the other state is 45,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 10 = 45$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 5 = 35$.

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humblest, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

We can now assign the cities and the NURs, the states, and the PM values uniquely based on our calculations. After assigning the table looks like,

CASE 1: Noodleton

If the other city of Humblest is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humblest.

CASE 2: Quackford

If the other city of Humblest is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humblest, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humblest.

CASE 3: Zingaloo

If the other city of Humblest is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

If the PI of the other state is 40,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

We can see that PI of the other two states is 40, which violates one of the conditions given.

If the PI of the other state is 45,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 10 = 45$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 5 = 35$.

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humblest, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

We can now assign the cities and the NURs the states and the PM values uniquely based on our calculations. After assigning the table looks like,

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
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Quackford	70	Foggia	17.5
Mumpypore	80	Whimshire	20
Zingaloo	90	Humbleset	22.5
NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humbleset)	40	Humbleset	20

We calculated the PI of Foggia to be 35.

Hence, the correct answer is 35.

[VIEW SOLUTION](#)

24. D

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humbleset. So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humbleset has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humbleset.

We can also conclude that the PMs of Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo are 50, 60, 70, 80, and 90, respectively.

We are given that in the PI calculation, cities have a contribution of 25% and NUR has a contribution of 50%. So, we can calculate the contribution of each city in the calculation of PI by dividing their PI value by 4, and the contribution of NUR can be calculated by dividing the PM value by 2.

CITY and NUR	PM	STATE	Contribution in PI
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Splutterville	60		15
Quackford	70		17.5
Mumpypore	80		20
Zingaloo	90		22.5
NUR(Whimshire)	20/10	Whimshire	10/5
NUR(Foggia)	10/20	Foggia	5/10
NUR(Humbleset)	40	Humbleset	20

We are given that all the PI values are unique integers, and Humbleset has the highest PI value, while Foggia has the lowest.

For the PI value to be an integer, the cities with decimal contributions must be paired with another city with decimal contributions. Since all the decimals are 0.5, when two cities with 0.5 are added, the sum becomes 1, making the PI an integer.

The cities Splutterville and Mumpypore must be from the same state; both have integral contributions towards the PI, and if they are added to a decimal, then the result will be a decimal, which should not be the case. These two cities have to belong to either the state of Whimshire or Foggia, but their combined contribution towards PI is $15 + 20 = 35$.

We already know that for the state of Humbleset, the PI is highest, and the two contributions towards the PI are 7.5 and 20, totalling 27.5.

There are three possibilities for the other city of Humbleset, which are Noodleton, Quackford and Zingaloo.

CASE 1: Noodleton

If the other city of Humblest is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humblest.

CASE 2: Quackford

If the other city of Humblest is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humblest, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humblest.

CASE 3: Zingaloo

If the other city of Humblest is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

If the PI of the other state is 40,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

We can see that PI of the other two states is 40, which violates one of the conditions given.

If the PI of the other state is 45,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 10 = 45$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 5 = 35$.

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humblest, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

We can now assign the cities and the NURs, the states, and the PM values uniquely based on our calculations. After assigning the table looks like,

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50	Foggia	12.5
Splutterville	60	Whimshire	15
Quackford	70	Foggia	17.5
Mumpypore	80	Whimshire	20
Zingaloo	90	Humbleset	22.5
NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humbleset)	40	Humbleset	20

We can see that Noodleton and Quackford belong to the same state among the options.

Hence, the correct answer is option C.

[VIEW SOLUTION](#)

25.9

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humbleset. So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humbleset has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humbleset.

We can also conclude that the PMs of Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo are 50, 60, 70, 80, and 90, respectively.

We are given that in the PI calculation, cities have a contribution of 25% and NUR has a contribution of 50%. So, we can calculate the contribution of each city in the calculation of PI by dividing their PI value by 4, and the contribution of NUR can be calculated by dividing the PM value by 2.

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50		12.5
Splutterville	60		15
Quackford	70		17.5
Mumpypore	80		20
Zingaloo	90		22.5
NUR(Whimshire)	20/10	Whimshire	10/5
NUR(Foggia)	10/20	Foggia	5/10
NUR(Humbleset)	40	Humbleset	20

We are given that all the PI values are unique integers, and Humbleset has the highest PI value, while Foggia has the lowest.

For the PI value to be an integer, the cities with decimal contributions must be paired with another city with decimal contributions. Since all the decimals are 0.5, when two cities with 0.5 are added, the sum becomes 1, making the PI an integer.

The cities Splutterville and Mumpypore must be from the same state; both have integral contributions towards the PI, and if they are added to a decimal, then the result will be a decimal, which should not be the case. These two cities have to belong to either the state of Whimshire or Foggia, but their combined contribution towards PI is $15 + 20 = 35$.

We already know that for the state of Humbleset, the PI is highest, and the two contributions towards the PI are 7.5 and 20, totalling 27.5.

There are three possibilities for the other city of Humbleset, which are Noodleton, Quackford and Zingaloo.

CASE 1: Noodleton

If the other city of Humblest is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humblest.

CASE 2: Quackford

If the other city of Humblest is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humblest, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humblest.

CASE 3: Zingaloo

If the other city of Humblest is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

If the PI of the other state is 40,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

We can see that PI of the other two states is 40, which violates one of the conditions given.

If the PI of the other state is 45,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 10 = 45$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 5 = 35$.

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humblest, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

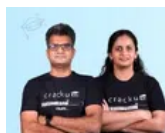
We can now assign the cities and the NURs, the states, and the PM values uniquely based on our calculations. After assigning the table looks like,

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50	Foggia	12.5
Splutterville	60	Whimshire	15
Quackford	70	Foggia	17.5
Mumpypore	80	Whimshire	20
Zingaloo	90	Humbleset	22.5
NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humbleset)	40	Humbleset	20

We can see that for all 6 cities and 3 NURs, the PMs and the city to which they belong can be identified uniquely. So, in total, we can identify the values uniquely for all 9 of them.

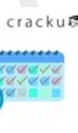
Hence, the correct answer is 9.

[VIEW SOLUTION](#)



CAT Daily Targets

Step by step, heading towards 99+ percentile



26. C

Peacock	is	designated	as	the	national	bird	of	India
5688999	35	1135556678	56	458	13666689	1334	79	13366

We can see that India's code is 13366 therefore we can say that I's code is either 3 or 6.

Also, we can see that code for word "is" is 35 therefore we can say that I's code is 3. Consequently, we can say that S's code is 5.

Also, we can see that code of word 'as' is 56 therefore we can say that A's code is 6. Consequently, we can say that S's code is 5.

There is only one letter 'O' common in words 'of' and 'national'. In code word as well only digit '9' is common in both. Hence, we can say that letter 'O' is assigned numerical '9'. Consequently, we can say that F is assigned number 7.

	1	2	3	4	5	6	7	8	9
Letters			I		S	A	F		O

It is given that '9' is assigned to only two alphabets one of them is 'O'. We can see that there are three 9's in Peacock's code. One of the digit '9' is used for 'O'. Remaining two 9's must represent same letter. We can see that only letter 'C' has appeared twice in Peacock. Therefore, we can say that 'C' is assigned number '9'.

In word national 'N' has appeared twice. In code only digit '6' has appeared more than once. Hence, we can say that code of letter N is '6'. Consequently, we can say that code for letter 'D' is '1' because in India rest of the numerals are already taken.

	1	2	3	4	5	6	7	8	9
Letters	D		I		S	N, A	F		C, O

In words, 'the' and 'national' only letter 't' is common. In code as well only digit '8' is common in two codes. Hence, we can say that letter code for letter 't' is 8.

In words, 'the' and 'peacock' only letter 'e' is common. In code as well only digit '5' is common in two codes. Hence, we can say that letter code for letter 'e' is 5. Consequently, we can say that leftover letter, in word "the", 'H's code is 4.

	1	2	3	4	5	6	7	8	9
Letters	D		I	H	S, E	N, A	F	T	C, O

We can see that code for word "NATIONAL" is 13666689. Hence, we can say that code for the letter L is '1'.

	1	2	3	4	5	6	7	8	9
Letters	L, D		I	H	S, E	N, A	F	T	C, O

We can see that code for word "DESIGNATED" is 1135556678. Hence, we can say that code for the letter 'G' is '7'.

	1	2	3	4	5	6	7	8	9
Letters	L, D		I	H	S, E	N, A	G, F	T	C, O

We can see that code for word "PEACOCK" is 5688999. Hence, we can say that code for the letters 'P' and 'K' is '8'.

	1	2	3	4	5	6	7	8	9
Letters	L, D		I	H	S, E	N, A	G, F	T, P, K	C, O

Let us check this by options:

- (A) S,E,Z: If letter 'Z' is assigned code '5' then this case is possible.
- (B) I,B,M: If letters 'B' and 'M' are assigned code '3' then this case is possible.
- (C) S,U,V: If letters 'U' and 'V' are assigned code '5' then this case is possible. But in that case digit 5 will have 4 letters associated with it which is not possible. Hence, this is the answer.
- (D) X,Y,Z: If letters 'X', 'Y' and 'Z' are assigned code '2' then this case is possible.



27.A

Peacock	is	designated	as	the	national	bird	of	India
5688999	35	1135556678	56	458	13666689	1334	79	13366

We can see that India's code is 13366 therefore we can say that I's code is either 3 or 6.

Also, we can see that code for word "is" is 35 therefore we can say that I's code is 3. Consequently, we can say that S's code is 5.

Also, we can see that code of word 'as' is 56 therefore we can say that A's code is 6. Consequently, we can say that S's code is 5.

There is only one letter 'O' common in words 'of' and 'national'. In code word as well only digit '9' is common in both. Hence, we can say that letter 'O' is assigned numerical '9'. Consequently, we can say that F is assigned number 7.

	1	2	3	4	5	6	7	8	9
Letters			I		S	A	F		O

It is given that '9' is assigned to only two alphabets one of them is 'O'. We can see that there are three 9's in Peacock's code. One of the digit '9' is used for 'O'. Remaining two 9's must represent same letter. We can see that only letter 'C' has appeared twice in Peacock. Therefore, we can say that 'C' is assigned number '9'.

In word national 'N' has appeared twice. In code only digit '6' has appeared more than once. Hence, we can say that code of letter N is '6'. Consequently, we can say that code for letter 'D' is '1' because in India rest of the numerals are already taken.

	1	2	3	4	5	6	7	8	9
Letters	D		I		S	N, A	F		C, O

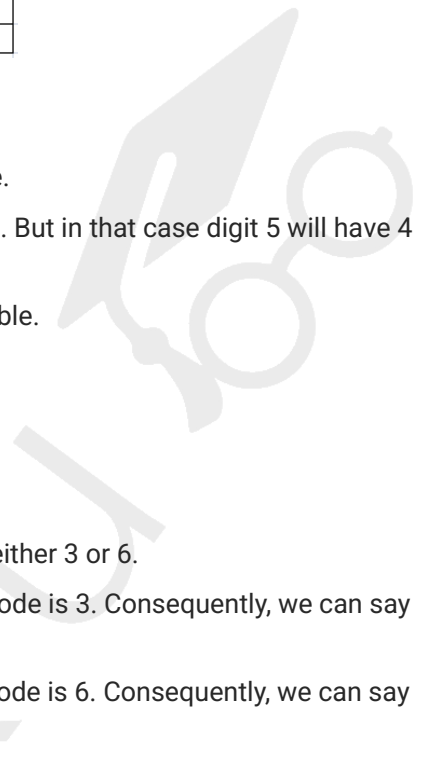
In words, 'the' and 'national' only letter 't' is common. In code as well only digit '8' is common in two codes. Hence, we can say that letter code for letter 't' is 8.

In words, 'the' and 'peacock' only letter 'e' is common. In code as well only digit '5' is common in two codes. Hence, we can say that letter code for letter 'e' is 5. Consequently, we can say that leftover letter, in word "the", 'H's code is 4.

	1	2	3	4	5	6	7	8	9
Letters	D		I	H	S. E	N. A	F	T	C. O

We can see that code for word "NATIONAL" is 13666689. Hence, we can say that code for the letter L is '1'.

Hence, option A is the correct answer.



28. A

Peacock	is	designated	as	the	national	bird	of	India
5688999	35	1135556678	56	458	13666689	1334	79	13366

We can see that India's code is 13366 therefore we can say that I's code is either 3 or 6.

Also, we can see that code for word "is" is 35 therefore we can say that I's code is 3. Consequently, we can say that S's code is 5.

Also, we can see that code of word 'as' is 56 therefore we can say that A's code is 6. Consequently, we can say that S's code is 5.

There is only one letter 'O' common in words 'of' and 'national'. In code word as well only digit '9' is common in both. Hence, we can say that letter 'O' is assigned numerical '9'. Consequently, we can say that F is assigned number 7.

	1	2	3	4	5	6	7	8	9
Letters			I		S	A	F		O

It is given that '9' is assigned to only two alphabets one of them is 'O'. We can see that there are three 9's in Peacock's code. One of the digit '9' is used for 'O'. Remaining two 9's must represent same letter. We can see that only letter 'C' has appeared twice in Peacock. Therefore, we can say that 'C' is assigned number '9'.

In word national 'N' has appeared twice. In code only digit '6' has appeared more than once. Hence, we can say that code of letter N is '6'. Consequently, we can say that code for letter 'D' is '1' because in India rest of the numerals are already taken.

	1	2	3	4	5	6	7	8	9
Letters	D		I		S	N, A	F		C, O

In words, 'the' and 'national' only letter 't' is common. In code as well only digit '8' is common in two codes. Hence, we can say that letter code for letter 't' is 8.

In words, 'the' and 'peacock' only letter 'e' is common. In code as well only digit '5' is common in two codes. Hence, we can say that letter code for letter 'e' is 5. Consequently, we can say that leftover letter, in word "the", 'H's code is 4.

	1	2	3	4	5	6	7	8	9
Letters	D		I	H	S, E	N, A	F	T	C, O

We can see that code for word "NATIONAL" is 13666689. Hence, we can say that code for the letter L is '1'.

	1	2	3	4	5	6	7	8	9
Letters	L, D		I	H	S, E	N, A	F	T	C, O

We can see that code for word "BIRD" is 1334. 1 corresponds to D and one 3 corresponds to I. Hence, we can say that code for letters 'R' and 'B' are '3' and '4' in any order.

Therefore, we can say that for letter 'B' there are two possible numbers: 3 or 4

Hence, option A is the correct answer.

 VIDEO SOLUTION

29. B

Peacock	is	designated	as	the	national	bird	of	India
5688999	35	1135556678	56	458	13666689	1334	79	13366

We can see that India's code is 13366 therefore we can say that I's code is either 3 or 6.

Also, we can see that code for word "is" is 35 therefore we can say that I's code is 3. Consequently, we can say that S's code is 5.

Also, we can see that code of word 'as' is 56 therefore we can say that A's code is 6. Consequently, we can say that S's code is 5.

There is only one letter 'O' common in words 'of' and 'national'. In code word as well only digit '9' is common in both. Hence, we can say that letter 'O' is assigned numerical '9'. Consequently, we can say that F is assigned number 7.

	1	2	3	4	5	6	7	8	9
Letters			I		S	A	F		O

It is given that '9' is assigned to only two alphabets one of them is 'O'. We can see that there are three 9's in Peacock's code. One of the digit '9' is used for 'O'. Remaining two 9's must represent same letter. We can see that only letter 'C' has appeared twice in Peacock. Therefore, we can say that 'C' is assigned number '9'.

In word national 'N' has appeared twice. In code only digit '6' has appeared more than once. Hence, we can say that code of letter N is '6'. Consequently, we can say that code for letter 'D' is '1' because in India rest of the numerals are already taken.

	1	2	3	4	5	6	7	8	9
Letters	D		I		S	N, A	F		C, O

In words, 'the' and 'national' only letter 't' is common. In code as well only digit '8' is common in two codes. Hence, we can say that letter code for letter 't' is 8.

In words, 'the' and 'peacock' only letter 'e' is common. In code as well only digit '5' is common in two codes. Hence, we can say that letter code for letter 'e' is 5. Consequently, we can say that leftover letter, in word "the", 'H's code is 4.

	1	2	3	4	5	6	7	8	9
Letters	D		I	H	S, E	N, A	F	T	C, O

We can see that code for word "NATIONAL" is 13666689. Hence, we can say that code for the letter L is '1'.

	1	2	3	4	5	6	7	8	9
Letters	L, D		I	H	S, E	N, A	F	T	C, O

We can see that code for word "DESIGNATED" is 1135556678. Hence, we can say that code for the letter 'G' is '7'.

	1	2	3	4	5	6	7	8	9
Letters	L, D		I	H	S, E	N, A	G, F	T	C, O

We can see that code for word "PEACOCK" is 5688999. Hence, we can say that code for the letters 'P' and 'K' is '8'.

	1	2	3	4	5	6	7	8	9
Letters	L, D		I	H	S, E	N, A	G, F	T, P, K	C, O

Digit '1' is used for L and D only. We can not figure out the third letter for which digit 1 is used.

Digit '2' is not used for any letter. Hence, we can not figure out all the letters for which digit 2 is correct code.

Digit '3' is used for letter 'I' only. Hence, we can not figure out all the letters for which digit 3 is correct code.

Digit '4' is used for letters 'H' and one of 'B' and 'R'. Hence, we can not figure out all the letters for which digit 4 is correct code.

Digit '5' is used for letters 'S' and 'E'. We can not figure out the third letter for which digit 5 is used.

Digit '6' is used for letters 'A' and 'N'. We can not figure out the third letter for which digit 6 is used.

Digit '7' is used for letters 'G' and 'F'. We can not figure out the third letter for which digit 7 is used.

Digit '8' is used for letters 'T', 'P' and K. Hence, we can say that this is one of the digit for which the complete list of letters associated is known.

Digit '9' is used for letters 'C' and 'O'. Hence, we can say that this is one of the digit for which the complete list of letters associated is known.

Therefore, we can say that for only two digits (8 and 9), the complete list of letters associated is known. Hence, option B is the correct answer.

 VIDEO SOLUTION

30. C

The patients in

Vial A: 9, 10, 11, 12, 13, 14, 15, 16

Vial B: 1, 2, 3, 4, 5, 6, 7, 8.

Vial C: 5,6,7,8,13,14,15,16

Vial D: 1,2,3,4,9,10,11,12

Vial E: 3,4,7,8,11,12,15,16

Vial F: 1,2,5,6,9,10,13,14

Vial G: 2,4,6,8,10,12,14,16

Vial H: 1,3,5,7,9,11,13,15

If vial C tests positive and vials A, E and H test negative then Patient 6 must have disease as all other patients in Vial C expect patient 6 are present in at least one of A, E, H.

 VIDEO SOLUTION



31. **A**

The patients in

Vial A: 9, 10, 11, 12, 13, 14, 15, 16

Vial B: 1, 2, 3, 4, 5, 6, 7, 8.

Vial C: 5,6,7,8,13,14,15,16

Vial D: 1,2,3,4,9,10,11,12

Vial E: 3,4,7,8,11,12,15,16

Vial F: 1,2,5,6,9,10,13,14

Vial G: 2,4,6,8,10,12,14,16

Vial H: 1,3,5,7,9,11,13,15

If vials C and D negative then no patient could test negative. Hence A is correct answer.

 VIDEO SOLUTION

32. **B**

The patients in

Vial A: 9, 10, 11, 12, 13, 14, 15, 16

Vial B: 1, 2, 3, 4, 5, 6, 7, 8.

Vial C: 5,6,7,8,13,14,15,16

Vial D: 1,2,3,4,9,10,11,12

Vial E: 3,4,7,8,11,12,15,16

Vial F: 1,2,5,6,9,10,13,14

Vial G: 2,4,6,8,10,12,14,16

Vial H: 1,3,5,7,9,11,13,15

Suppose vial A tests positive and vials D and G test negative then the patient who tested positive must be one of patient 13 or 15.

Patient 13 or 15 are not present in vial B. So, A is not the answer.

Both patients present in vial C. Even if tested positive or negative we can't know who has got the disease. So, C is not the answer.

Both patients present in vial H. Even if tested positive or negative we can't know who has got the disease. So, H is not the answer.

only patient 15 is present in vial E, if tested positive then patient 15 has the disease else patient 13 as disease.

Hence Option 2 is correct.

 VIDEO SOLUTION

33. C

Let one of the patients, patient 1 or patient 16 has the disease and his blood is mixed with other then all 8 vials will test positive. $\Rightarrow$

8 has to be one of the answers.

If patient 2 and patient 16's blood is mixed and one of them has the disease then 7 of the 8 vials will test positive. So 7 has to be there in the option.

If 1 has the disease and 1, 7 are mixed then 6 out of the 8 vials test positive.

If 1 has the disease and 1, 9 are mixed then 5 of the 8 vials test positive,

Now, let us assume that patient 1 has the disease if his blood is not mixed, then 4 vials will definitely show positive.

Hence 3 is the correct answer.

 VIDEO SOLUTION

34. A

From statement 6, it can be concluded that the total number of new cases is equal to $12+12+5+14 = 43$.

Also, since the total number of cases in Kitmisto is 14, it can be concluded that the number of cases each day is either 2 or 3, where 3 cases are recorded on 4 days and 2 cases are recorded on 1 day.

From statement 4, it can be concluded that the number of new cases for Pismisto will be 0, 1, 1, 1, and 2, in any order (Since the total number of cases is 5, and the maximum number of new cases is 2).

In statement 3, it is given that the number of new cases kept increasing during the 5-day period.

Now, as it is already known that the maximum number of cases for Pismisto is 2, the maximum total number of cases in a day (or on day 5) will be less than 12.

Let us consider the maximum number of cases on Day 5 as 10.

Thus the maximum number of cases possible for the remaining days will be 9, 8, 7, and 6. So, the total number of maximum cases possible for this case will be 40 (less than 43).

Thus, the number of cases on Day 5 will be 11 (i.e., b/w 10 and 11)

Now, if the number of cases on Day 4 is 9, the maximum number of cases possible for the remaining days will be 8, 7, and 6.

Thus, the maximum number of cases, in this case, will be 41 (less than 43).

So, the number of cases on day 4 will be 10.

Now, if the number of cases on Day 3 is 8, the number of cases on day 2 will be 7, and the maximum possible number of cases on Day 1 will be 6.

Thus, the number of cases, in this case, will be 42 (less than 43).

Thus, the number of cases on day 3 will be 9, the number of cases on day 2 will be 8, and the number of cases on day 1 will be 5.

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto						12
Tyhrmisto						12
Pesmisto						5
Kitmisto						14
Total	5	8	9	10	11	43

Since all the neighbourhood has at least one case on Day 1, the only possible combination will be 1, 1, 1, and 2 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, for the other 4 days, the number of cases in Kitmisto will be 3.

For day 5, the number of cases will be 3, 3, 2, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively(since the maximum number of cases in Pesmisto is 2).

And since Pesmisto only has the maximum number of cases on one day, the number of cases on day 4 will be 3, 3, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

On day 2, since Kitmisto is the only neighbourhood to have 3 cases, the number of cases on day 2 will be 2, 2, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, on day 3, the number of cases will be 3, 3, 0, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Thus, the final table will be as follows:

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto	1	2	3	3	3	12
Tyhrmisto	1	2	3	3	3	12
Pesmisto	1	1	0	1	2	5
Kitmisto	2	3	3	3	3	14
Total	5	8	9	10	11	43

From the final table, it can be concluded that on Day 3, the number of cases will be zero for Pesmisto.

Thus, the correct option is A.

 VIDEO SOLUTION

35. D

From statement 6, it can be concluded that the total number of new cases is equal to $12+12+5+14 = 43$.

Also, since the total number of cases in Kitmisto is 14, it can be concluded that the number of cases each day is either 2 or 3, where 3 cases are recorded on 4 days and 2 cases are recorded on 1 day.

From statement 4, it can be concluded that the number of new cases for Pesmisto will be 0,1,1,1, and 2, in any order(Since the total number of cases is 5, and the maximum number of new cases is 2).

In statement 3, it is given that the number of new cases kept increasing during the 5-day period.

Now, as it is already known that the maximum number of cases for Pesmisto is 2, the maximum total number of cases in a day(or on day 5) will be less than 12.

Let us consider the maximum number of cases on Day 5 as 10.

Thus the maximum number of cases possible for the remaining days will be 9, 8, 7, and 6. So, the total number of maximum cases possible for this case will be 40 (less than 43).

Thus, the number of cases on Day 5 will be 11 (i.e., b/w 10 and 11)

Now, if the number of cases on Day 4 is 9, the maximum number of cases possible for the remaining days will be 8, 7, and 6.

Thus, the maximum number of cases, in this case, will be 41 (less than 43).

So, the number of cases on day 4 will be 10.

Now, if the number of cases on Day 3 is 8, the number of cases on day 2 will be 7, and the maximum possible number of cases on Day 1 will be 6.

Thus, the number of cases, in this case, will be 42 (less than 43).

Thus, the number of cases on day 3 will be 9, the number of cases on day 2 will be 8, and the number of cases on day 1 will be 5.

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto						12
Tyhrmisto						12
Pesmisto						5
Kitmisto						14
Total	5	8	9	10	11	43

Since all the neighbourhood has at least one case on Day 1, the only possible combination will be 1, 1, 1, and 2 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, for the other 4 days, the number of cases in Kitmisto will be 3.

For day 5, the number of cases will be 3, 3, 2, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively (since the maximum number of cases in Pesmisto is 2).

And since Pesmisto only has the maximum number of cases on one day, the number of cases on day 4 will be 3, 3, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

On day 2, since Kitmisto is the only neighbourhood to have 3 cases, the number of cases on day 2 will be 2, 2, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, on day 3, the number of cases will be 3, 3, 0, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Thus, the final table will be as follows:

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto	1	2	3	3	3	12
Tyhrmisto	1	2	3	3	3	12
Pesmisto	1	1	0	1	2	5
Kitmisto	2	3	3	3	3	14
Total	5	8	9	10	11	43

From the data, it can be concluded that the total number of cases on Day 2 is equal to 8.

Thus, the correct option is D.



36. C

From statement 6, it can be concluded that the total number of new cases is equal to $12+12+5+14 = 43$.

Also, since the total number of cases in Kitmisto is 14, it can be concluded that the number of cases each day is either 2 or 3, where 3 cases are recorded on 4 days and 2 cases are recorded on 1 day.

From statement 4, it can be concluded that the number of new cases for Pesmisto will be 0,1,1,1, and 2, in any order(Since the total number of cases is 5, and the maximum number of new cases is 2).

In statement 3, it is given that the number of new cases kept increasing during the 5-day period.

Now, as it is already known that the maximum number of cases for Pesmisto is 2, the maximum total number of cases in a day(or on day 5) will be less than 12.

Let us consider the maximum number of cases on Day 5 as 10.

Thus the maximum number of cases possible for the remaining days will be 9, 8, 7, and 6. So, the total number of maximum cases possible for this case will be 40(less than 43).

Thus, the number of cases on Day 5 will be 11(i.e., b/w 10 and 11)

Now, if the number of cases on Day 4 is 9, the maximum number of cases possible for the remaining days will be 8, 7, and 6.

Thus, the maximum number of cases, in this case, will be 41(less than 43).

So, the number of cases on day 4 will be 10.

Now, if the number of cases on Day 3 is 8, the number of cases on day 2 will be 7, and the maximum possible number of cases on Day 1 will be 6.

Thus, the number of cases, in this case, will be 42(less than 43).

Thus, the number of cases on day 3 will be 9, the number of cases on day 2 will be 8, and the number of cases on day 1 will be 5.

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto						12
Tyhrmisto						12
Pesmisto						5
Kitmisto						14
Total	5	8	9	10	11	43

Since all the neighbourhood has at least one case on Day 1, the only possible combination will be 1, 1, 1, and 2 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, for the other 4 days, the number of cases in Kitmisto will be 3.

For day 5, the number of cases will be 3, 3, 2, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively(since the maximum number of cases in Pesmisto is 2).

And since Pesmisto only has the maximum number of cases on one day, the number of cases on day 4 will be 3, 3, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

On day 2, since Kitmisto is the only neighbourhood to have 3 cases, the number of cases on day 2 will be 2, 2, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, on day 3, the number of cases will be 3, 3, 0, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Thus, the final table will be as follows:

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto	1	2	3	3	3	12
Tyhrmisto	1	2	3	3	3	12
Pesmisto	1	1	0	1	2	5
Kitmisto	2	3	3	3	3	14
Total	5	8	9	10	11	43

From the final table, it can be concluded that the total number of cases in Levmosto is 3 on day 3.

Thus, the correct option is C.

 VIDEO SOLUTION

37. D

From statement 6, it can be concluded that the total number of new cases is equal to $12+12+5+14 = 43$.

Also, since the total number of cases in Kitmisto is 14, it can be concluded that the number of cases each day is either 2 or 3, where 3 cases are recorded on 4 days and 2 cases are recorded on 1 day.

From statement 4, it can be concluded that the number of new cases for Pesmisto will be 0,1,1,1, and 2, in any order(Since the total number of cases is 5, and the maximum number of new cases is 2).

In statement 3, it is given that the number of new cases kept increasing during the 5-day period.

Now, as it is already known that the maximum number of cases for Pesmisto is 2, the maximum total number of cases in a day(or on day 5) will be less than 12.

Let us consider the maximum number of cases on Day 5 as 10.

Thus the maximum number of cases possible for the remaining days will be 9, 8, 7, and 6. So, the total number of maximum cases possible for this case will be 40(less than 43).

Thus, the number of cases on Day 5 will be 11(i.e., b/w 10 and 11)

Now, if the number of cases on Day 4 is 9, the maximum number of cases possible for the remaining days will be 8, 7, and 6.

Thus, the maximum number of cases, in this case, will be 41(less than 43).

So, the number of cases on day 4 will be 10.

Now, if the number of cases on Day 3 is 8, the number of cases on day 2 will be 7, and the maximum possible number of cases on Day 1 will be 6.

Thus, the number of cases, in this case, will be 42(less than 43).

Thus, the number of cases on day 3 will be 9, the number of cases on day 2 will be 8, and the number of cases on day 1 will be 5.

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto						12
Tyhrmisto						12
Pesmisto						5
Kitmisto						14
Total	5	8	9	10	11	43

Since all the neighbourhood has at least one case on Day 1, the only possible combination will be 1, 1, 1, and 2 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, for the other 4 days, the number of cases in Kitmisto will be 3.

For day 5, the number of cases will be 3, 3, 2, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively(since the maximum number of cases in Pesmisto is 2).

And since Pesmisto only has the maximum number of cases on one day, the number of cases on day 4 will be 3, 3, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

On day 2, since Kitmisto is the only neighbourhood to have 3 cases, the number of cases on day 2 will be 2, 2, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, on day 3, the number of cases will be 3, 3, 0, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Thus, the final table will be as follows:

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto	1	2	3	3	3	12
Tyhrmisto	1	2	3	3	3	12
Pesmisto	1	1	0	1	2	5
Kitmisto	2	3	3	3	3	14
Total	5	8	9	10	11	43

It can be concluded from the final table that the number of cases will be the same for all the days.

Thus, the correct option is D

 VIDEO SOLUTION

38. D

From statement 6, it can be concluded that the total number of new cases is equal to $12+12+5+14 = 43$.

Also, since the total number of cases in Kitmisto is 14, it can be concluded that the number of cases each day is either 2 or 3, where 3 cases are recorded on 4 days and 2 cases are recorded on 1 day.

From statement 4, it can be concluded that the number of new cases for Pesmisto will be 0,1,1,1, and 2, in any order(Since the total number of cases is 5, and the maximum number of new cases is 2).

In statement 3, it is given that the number of new cases kept increasing during the 5-day period.

Now, as it is already known that the maximum number of cases for Pesmisto is 2, the maximum total number of cases in a day(or on day 5) will be less than 12.

Let us consider the maximum number of cases on Day 5 as 10.

Thus the maximum number of cases possible for the remaining days will be 9, 8, 7, and 6. So, the total number of maximum cases possible for this case will be 40 (less than 43).

Thus, the number of cases on Day 5 will be 11 (i.e., b/w 10 and 11)

Now, if the number of cases on Day 4 is 9, the maximum number of cases possible for the remaining days will be 8, 7, and 6.

Thus, the maximum number of cases, in this case, will be 41 (less than 43).

So, the number of cases on day 4 will be 10.

Now, if the number of cases on Day 3 is 8, the number of cases on day 2 will be 7, and the maximum possible number of cases on Day 1 will be 6.

Thus, the number of cases, in this case, will be 42 (less than 43).

Thus, the number of cases on day 3 will be 9, the number of cases on day 2 will be 8, and the number of cases on day 1 will be 5.

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto						12
Tyhrmisto						12
Pesmisto						5
Kitmisto						14
Total	5	8	9	10	11	43

Since all the neighbourhood has at least one case on Day 1, the only possible combination will be 1, 1, 1, and 2 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, for the other 4 days, the number of cases in Kitmisto will be 3.

For day 5, the number of cases will be 3, 3, 2, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively (since the maximum number of cases in Pesmisto is 2).

And since Pesmisto only has the maximum number of cases on one day, the number of cases on day 4 will be 3, 3, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

On day 2, since Kitmisto is the only neighbourhood to have 3 cases, the number of cases on day 2 will be 2, 2, 1, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, on day 3, the number of cases will be 3, 3, 0, and 3 for Levmosto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Thus, the final table will be as follows:

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto	1	2	3	3	3	12
Tyhrmisto	1	2	3	3	3	12
Pesmisto	1	1	0	1	2	5
Kitmisto	2	3	3	3	3	14
Total	5	8	9	10	11	43

From the final table, it can be concluded that both statements are false.

Thus the correct option is D.

39. A

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	
210	2,3,5,7	
165	3,5,11	
77	7,11	
66	2,3,11	

From statement 1 we can deduce that Fatima gave token 3 and Adhara gave token 13 and therefore from this we can say that Qahira received a funding of 77,000 and Pragnyaa received a funding of 390,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000.

Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	
77	7,11	Qahira
66	2,3,11	

From statement 3 we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	Smera/Tantra
77	7,11	Qahira
66	2,3,11	Smera/Tantra

Person	Token
Adhara	13
Bithi	2/5
Chhaya	5/2
Dhanavi	7
Esther	11
Fathima	3

Alternate Explanation:

$$390 = 2 \times 3 \times 5 \times 13$$

$$210 = 2 \times 3 \times 5 \times 7$$

$$165 = 3 \times 5 \times 11$$

$$77 = 7 \times 11$$

$$66 = 2 \times 3 \times 11$$

From the above information we can conclude that the number of times a particular token was given, therefore we get the following table,

Token	No. of times given
2	3
3	4
5	3
7	2
11	2
13	1

We know that there are five people who received the token and there are 6 people who awarded the token.

From, statement 1 we know that Fatima gave token to 4 people except Qahira so the token number given by Fatima is 3, and Adhara gave token only to Pragnyaa so the token number given by Adhara is 13. Therefore we can also say that Pragnyaa received 390,000 and Qahira received 77,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000. Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

	Pragnyaa (390)	Qahira (77)	Rashida (210)	Smera	Tantra
Adhara(13)	✓	X	X	X	X
Bithi			✓		
Chhaya			✓		
Dhanavi			✓		
Esther(11)	X	✓	X	✓	✓
Fathima(3)	✓	X	✓	✓	✓

From statement 3, we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

	Pragnyaa (390)	Qahira (77)	Rashida (210)	Smera (66/165)	Tantra (165/66)
Adhara (13)	✓	X	X	X	X
Bithi (2/5)	✓	X	✓		
Chhaya (2/5)	✓	X	✓		
Dhanavi (7)	X	✓	✓	X	X
Esther (11)	X	✓	X	✓	✓
Fathima (3)	✓	X	✓	✓	✓

Ans:A

[VIDEO SOLUTION](#)

40.3

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	
210	2,3,5,7	
165	3,5,11	
77	7,11	
66	2,3,11	

From statement 1 we can deduce that Fatima gave token 3 and Adhara gave token 13 and therefore from this we can say that Qahira received a funding of 77,000 and Pragnyaa received a funding of 390,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000.

Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	
77	7,11	Qahira
66	2,3,11	

From statement 3 we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	Smera/Tantra
77	7,11	Qahira
66	2,3,11	Smera/Tantra

Person	Token
Adhara	13
Bithi	2/5
Chhaya	5/2
Dhanavi	7
Esther	11
Fathima	3

Alternate Explanation:

$$390 = 2 \times 3 \times 5 \times 13$$

$$210 = 2 \times 3 \times 5 \times 7$$

$$165 = 3 \times 5 \times 11$$

$$77 = 7 \times 11$$

$$66 = 2 \times 3 \times 11$$

From the above information we can conclude that the number of times a particular token was given, therefore we get the following table,

Token	No. of times given
2	3
3	4
5	3
7	2
11	2
13	1

We know that there are five people who received the token and there are 6 people who awarded the token.

From, statement 1 we know that Fatima gave token to 4 people except Qahira so the token number given by Fatima is 3, and Adhara gave token only to Pragnyaa so the token number given by Adhara is 13. Therefore we can also say that Pragnyaa received 390,000 and Qahira received 77,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000. Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

	Pragnyaa (390)	Qahira (77)	Rashida (210)	Smera	Tantra
Adhara(13)	✓	X	X	X	X
Bithi			✓		
Chhaya			✓		
Dhanavi			✓		
Esther(11)	X	✓	X	✓	✓
Fathima(3)	✓	X	✓	✓	✓

From statement 3, we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

	Pragnyaa (390)	Qahira (77)	Rashida (210)	Smera (66/165)	Tantra (165/66)
Adhara (13)	✓	X	X	X	X
Bithi (2/5)	✓	X	✓		
Chhaya (2/5)	✓	X	✓		
Dhanavi (7)	X	✓	✓	X	X
Esther (11)	X	✓	X	✓	✓
Fathima (3)	✓	X	✓	✓	✓

Ans:3



41.3

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	
210	2,3,5,7	
165	3,5,11	
77	7,11	
66	2,3,11	

From statement 1 we can deduce that Fatima gave token 3 and Adhara gave token 13 and therefore from this we can say that Qahira received a funding of 77,000 and Pragnyaa received a funding of 390,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000.

Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	
77	7,11	Qahira
66	2,3,11	

From statement 3 we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	Smera/Tantra
77	7,11	Qahira
66	2,3,11	Smera/Tantra

Person	Token
Adhara	13
Biti	2/5
Chhaya	5/2
Dhanavi	7
Esther	11
Fathima	3

Alternate Explanation:

$$390 = 2 \times 3 \times 5 \times 13$$

$$210 = 2 \times 3 \times 5 \times 7$$

$$165 = 3 \times 5 \times 11$$

$$77 = 7 \times 11$$

$$66 = 2 \times 3 \times 11$$

From the above information we can conclude that the number of times a particular token was given, therefore we get the following table,

Token	No. of times given
2	3
3	4
5	3
7	2
11	2
13	1

We know that there are five people who received the token and there are 6 people who awarded the token.

From, statement 1 we know that Fatima gave token to 4 people except Qahira so the token number given by Fatima is 3, and Adhara gave token only to Pragnyaa so the token number given by Adhara is 13. Therefore we can also say that Pragnyaa received 390,000 and Qahira received 77,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000. Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

	Pragnyaa (390)	Qahira (77)	Rashida (210)	Smera	Tantra
Adhara(13)	✓	X	X	X	X
Bithi			✓		
Chhaya			✓		
Dhanavi			✓		
Esther(11)	X	✓	X	✓	✓
Fathima(3)	✓	X	✓	✓	✓

From statement 3, we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

	Pragnyaa (390)	Qahira (77)	Rashida (210)	Smera (66/165)	Tantra (165/66)
Adhara (13)	✓	X	X	X	X
Bithi (2/5)	✓	X	✓		
Chhaya (2/5)	✓	X	✓		
Dhanavi (7)	X	✓	✓	X	X
Esther (11)	X	✓	X	✓	✓
Fathima (3)	✓	X	✓	✓	✓

Ans:3

[VIDEO SOLUTION](#)

42. D

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	
210	2,3,5,7	
165	3,5,11	
77	7,11	
66	2,3,11	

From statement 1 we can deduce that Fatima gave token 3 and Adhara gave token 13 and therefore from this we can say that Qahira received a funding of 77,000 and Pragnyaa received a funding of 390,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000.

Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	
77	7,11	Qahira
66	2,3,11	

From statement 3 we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	Smera/Tantra
77	7,11	Qahira
66	2,3,11	Smera/Tantra

Person	Token
Adhara	13
Bithi	2/5
Chhaya	5/2
Dhanavi	7
Esther	11
Fathima	3

Alternate Explanation:

$$390 = 2 \times 3 \times 5 \times 13$$

$$210 = 2 \times 3 \times 5 \times 7$$

$$165 = 3 \times 5 \times 11$$

$$77 = 7 \times 11$$

$$66 = 2 \times 3 \times 11$$

From the above information we can conclude that the number of times a particular token was given, therefore we get the following table,

Token	No. of times given
2	3
3	4
5	3
7	2
11	2
13	1

We know that there are five people who received the token and there are 6 people who awarded the token.

From, statement 1 we know that Fatima gave token to 4 people except Qahira so the token number given by Fatima is 3, and Adhara gave token only to Pragnyaa so the token number given by Adhara is 13. Therefore we can also say that Pragnyaa received 390,000 and Qahira received 77,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000. Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

	Pragnyaa (390)	Qahira (77)	Rashida (210)	Smera	Tantra
Adhara(13)	✓	X	X	X	X
Bithi			✓		
Chhaya			✓		
Dhanavi			✓		
Esther(11)	X	✓	X	✓	✓
Fathima(3)	✓	X	✓	✓	✓

From statement 3, we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

	Pragnyaa (390)	Qahira (77)	Rashida (210)	Smera (66/165)	Tantra (165/66)
Adhara (13)	✓	X	X	X	X
Bithi (2/5)	✓	X	✓		
Chhaya (2/5)	✓	X	✓		
Dhanavi (7)	X	✓	✓	X	X
Esther (11)	X	✓	X	✓	✓
Fathima (3)	✓	X	✓	✓	✓

Ans:D

43.2

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	
210	2,3,5,7	
165	3,5,11	
77	7,11	
66	2,3,11	

From statement 1 we can deduce that Fatima gave token 3 and Adhara gave token 13 and therefore from this we can say that Qahira received a funding of 77,000 and Pragnyaa received a funding of 390,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000.

Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	
77	7,11	Qahira
66	2,3,11	

From statement 3 we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

Amount(in 000s)	Tokens Received	Person
390	2,3,5,13	Pragnyaa
210	2,3,5,7	Rashida
165	3,5,11	Smera/Tantra
77	7,11	Qahira
66	2,3,11	Smera/Tantra

Person	Token
Adhara	13
Bithi	2/5
Chhaya	5/2
Dhanavi	7
Esther	11
Fathima	3

Alternate Explanation:

$$390 = 2 \times 3 \times 5 \times 13$$

$$210 = 2 \times 3 \times 5 \times 7$$

$$165 = 3 \times 5 \times 11$$

$$77 = 7 \times 11$$

$$66 = 2 \times 3 \times 11$$

From the above information we can conclude that the number of times a particular token was given, therefore we get the following table,

Token	No. of times given
2	3
3	4
5	3
7	2
11	2
13	1

We know that there are five people who received the token and there are 6 people who awarded the token.

From, statement 1 we know that Fatima gave token to 4 people except Qahira so the token number given by Fatima is 3, and Adhara gave token only to Pragnyaa so the token number given by Adhara is 13. Therefore we can also say that Pragnyaa received 390,000 and Qahira received 77,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000. Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan			Barun				
Major	Finance			Ops		Ops		
Minor				Finance	Finance			

It has been given that one female student and 2 male students major in finance. We know that the male student from Y minors in finance. Therefore, he cannot major in finance. Therefore, both the male students from X should major in finance.

Daisy and Amit are from the same institute. Therefore, Daisy cannot be from institute Z (since Amit is a male student and both the students from Z are female). Daisy minors in operations. The girl from institute Y majors in Operations. Therefore, Daisy cannot be from institute Y as well. Daisy and Amit should be from institute X. 3 female students minor in marketing. Therefore, all girls except Daisy should minor in marketing.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan	Amit	Daisy	Barun				
Major	Finance	Finance		Ops		Ops		
Minor			Ops	Finance	Finance	Marketing	Marketing	Marketing

Adriana and Deb are from the same institute. Therefore, both of them should be from institute Y. Bandita and Chitra should be from institute Z.

Only one male student majors in Operations. We know that Barun is the student. Two male students major in Finance. We know that Amit and Chetan major in finance. Therefore, Deb should major in Marketing.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan	Amit	Daisy	Barun	Deb	Adriana	Bandita	Chitra
Major	Finance	Finance		Ops	Marketing	Ops		
Minor			Ops	Finance	Finance	Marketing	Marketing	Marketing

Deb minors in Finance. Therefore, option B is the right answer.

[VIDEO SOLUTION](#)

45. D

There are 8 students in total - 4 male and 4 female. There are 3 institutes X, Y, and Z.

3 students are from institute X, 3 students are from institute Y, and 2 students are from institute Z. No student majors and minors in the same subject.

It has been given that both the students from institute Z are female. Also, it has been given that both the male students from institute Y minor in Finance. Therefore, the third student from institute Y should be female. Institute X should also have 2 male and 1 female student.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name								
Major								
Minor								

Both the male students from Y minor in Finance, while the female student from Y majors in Operations. Barun is from Y and majors in Operations. Chetan is from X and majors in Finance.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan			Barun				
Major	Finance			Ops		Ops		
Minor				Finance	Finance			

It has been given that one female student and 2 male students major in finance. We know that the male student from Y minors in finance. Therefore, he cannot major in finance. Therefore, both the male students from X should major in finance.

Daisy and Amit are from the same institute. Therefore, Daisy cannot be from institute Z (since Amit is a male student and both the students from Z are female). Daisy minors in operations. The girl from institute Y majors in Operations. Therefore, Daisy cannot be from institute Y as well. Daisy and Amit should be from institute X. 3 female students minor in marketing. Therefore, all girls except Daisy should minor in marketing.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan	Amit	Daisy	Barun				
Major	Finance	Finance		Ops		Ops		
Minor			Ops	Finance	Finance	Marketing	Marketing	Marketing

Adriana and Deb are from the same institute. Therefore, both of them should be from institute Y. Bandita and Chitra should be from institute Z.

Only one male student majors in Operations. We know that Barun is the student. Two male students major in Finance. We know that Amit and Chetan major in finance. Therefore, Deb should major in Marketing.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan	Amit	Daisy	Barun	Deb	Adriana	Bandita	Chitra
Major	Finance	Finance		Ops	Marketing	Ops		
Minor			Ops	Finance	Finance	Marketing	Marketing	Marketing

Amit majors in finance. Therefore, option D is the right answer.

VIDEO SOLUTION

46. C

There are 8 students in total - 4 male and 4 female. There are 3 institutes X, Y, and Z.

3 students are from institute X, 3 students are from institute Y, and 2 students are from institute Z. No student majors and minors in the same subject.

It has been given that both the students from institute Z are female. Also, it has been given that both the male students from institute Y minor in Finance. Therefore, the third student from institute Y should be female. Institute X should also have 2 male and 1 female student.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name								
Major								
Minor								

Both the male students from Y minor in Finance, while the female student from Y majors in Operations. Barun is from Y and majors in Operations. Chetan is from X and majors in Finance.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan			Barun				
Major	Finance			Ops		Ops		
Minor				Finance	Finance			

It has been given that one female student and 2 male students major in finance. We know that the male student from Y minors in finance. Therefore, he cannot major in finance. Therefore, both the male students from X should major in finance.

Daisy and Amit are from the same institute. Therefore, Daisy cannot be from institute Z (since Amit is a male student and both the students from Z are female). Daisy minors in operations. The girl from institute Y majors in Operations. Therefore, Daisy cannot be from institute Y as well. Daisy and Amit should be from institute X. 3 female students minor in marketing. Therefore, all girls except Daisy should minor in marketing.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan	Amit	Daisy	Barun				
Major	Finance	Finance		Ops		Ops		
Minor			Ops	Finance	Finance	Marketing	Marketing	Marketing

Adriana and Deb are from the same institute. Therefore, both of them should be from institute Y. Bandita and Chitra should be from institute Z.

Only one male student majors in Operations. We know that Barun is the student. Two male students major in Finance. We know that Amit and Chetan major in finance. Therefore, Deb should major in Marketing.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan	Amit	Daisy	Barun	Deb	Adriana	Bandita	Chitra
Major	Finance	Finance		Ops	Marketing	Ops		
Minor			Ops	Finance	Finance	Marketing	Marketing	Marketing

If Chitra majors in finance, Bandita cannot major in finance (only one female student majors in finance). She cannot major in marketing as well (since she has a minor degree in marketing). Therefore, Bandita should major in operations and hence, option C is the right answer.

 VIDEO SOLUTION

47. C

There are 8 students in total - 4 male and 4 female. There are 3 institutes X, Y, and Z.

3 students are from institute X, 3 students are from institute Y, and 2 students are from institute Z. No student majors and minors in the same subject.

It has been given that both the students from institute Z are female. Also, it has been given that both the male students from institute Y minor in Finance. Therefore, the third student from institute Y should be female. Institute X should also have 2 male and 1 female student.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name								
Major								
Minor								

Both the male students from Y minor in Finance, while the female student from Y majors in Operations. Barun is from Y and majors in Operations. Chetan is from X and majors in Finance.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan			Barun				
Major	Finance			Ops		Ops		
Minor				Finance	Finance			

It has been given that one female student and 2 male students major in finance. We know that the male student from Y minors in finance. Therefore, he cannot major in finance. Therefore, both the male students from X should major in finance.

Daisy and Amit are from the same institute. Therefore, Daisy cannot be from institute Z (since Amit is a male student and both the students from Z are female). Daisy minors in operations. The girl from institute Y majors in Operations. Therefore, Daisy cannot be from institute Y as well. Daisy and Amit should be from institute X. 3 female students minor in marketing. Therefore, all girls except Daisy should minor in marketing.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan	Amit	Daisy	Barun				
Major	Finance	Finance		Ops		Ops		
Minor			Ops	Finance	Finance	Marketing	Marketing	Marketing

Adriana and Deb are from the same institute. Therefore, both of them should be from institute Y. Bandita and Chitra should be from institute Z.

Only one male student majors in Operations. We know that Barun is the student. Two male students major in Finance. We know that Amit and Chetan major in finance. Therefore, Deb should major in Marketing.

College	X			Y			Z	
Gender	M	M	F	M	M	F	F	F
Name	Chetan	Amit	Daisy	Barun	Deb	Adriana	Bandita	Chitra
Major	Finance	Finance		Ops	Marketing	Ops		
Minor			Ops	Finance	Finance	Marketing	Marketing	Marketing

Bandita and Chitra are from institute Z. Therefore, option C is the right answer.

VIDEO SOLUTION

48. D

Now, we know there is only 1 candidate from OQ, which means that the number of non-candidate voters in OQ will be 4.

We also know that the non-candidates in a particular department voted as a block, and we also know that the least number of non-candidate voters in a particular department can be 1 (BH, 3-2 faculty).

Now, we also know that R got 5 votes from non-candidates.

Now we can write 5 as

i) 5

ii) 4+1

iii) 3+2

Considering case (i) 4+1. This is only possible when there is 1 candidate from OQ, and there are 2 candidates from BH. This implies that the number of candidates in FA and MQ is 1. Now, if we consider FA and MQ and put only 1 candidate there, it implies that there are 15 non-candidate voters between them. Now we know this is not possible since the maximum number of non-candidate voters a candidate can get is 13. (Please note that non-candidates of a particular department vote as a block).

On similar grounds, we can eliminate Case (iii) as it also implies there is only 1 candidate in FA and MQ.

Now, considering Case (i), we know that 5+0 will happen only one when there are 5 non-candidates in a single department. This is only possible in MS (Out of 7, there will be 2 candidates and 5 non-candidates).

So we can conclude that MS has 2 candidates and that they voted Prof. R.....(i)

We also know that Prof P got 2 votes from Non-candidates. This is only possible when BH has 1 candidate.

So, we can conclude that the number of professors in FA, MS, OQ, BH is 0,2,1,1

	FA	MS	OQ	BH
Total number of voters	9	7	5	3
Number of candidates	0	2	1	1
Number of non-candidates	9	5	4	2

Thus, we get the following table:

	P	Q	R	S
Total Votes	3	14	6	1
Candidate Vote	1(S)	1(R)	1(P)	1(Q)
Votes from non candidates	2	13	5	0
	BH	FA+OQ	MS	

Now, if we consider Department MS, we know that there are 2 candidates from MS and R can't be one of them as the people in that department voted for him..... (3 rd condition).

So the possible combinations of candidates in MS are (P,Q), (Q,S), (P,S).

Now we also know that no one can vote for a candidate in their own department, so we can eliminate (P,S) and (Q,S) as we know that S voted for P and Q voted for S).

So we can infer that P and Q are from MS.

	FA	MS	OQ	BH
Case 1	0	P, Q	R	S
Case 2	0	P, Q	S	R

Now, we can see that the number of votes that Prof Qureshi received from a single department can be 9 or 5 (if R is from OQ) or 4 (if R is not from OQ).

So, among the options, only Option D can be true. Therefore, Option D is the correct answer.

[VIDEO SOLUTION](#)

49. A

Now, we know there is only 1 candidate from OQ, which means that the number of non-candidate voters in OQ will be 4.

We also know that the non-candidates in a particular department voted as a block, and we also know that the least number of non-candidate voters in a particular department can be 1 (BH, 3-2 faculty).

Now, we also know that R got 5 votes from non-candidates.

Now we can write 5 as

i)5

ii)4+1

iii)3+2

Considering case (i) 4+1. This is only possible when there is 1 candidate from OQ, and there are 2 candidates from BH. This implies that the number of candidates in FA and MQ is 1. Now, if we consider FA and MQ and put only 1 candidate there, it implies that there are 15 non-candidate voters between them. Now we know this is not possible since the maximum number of non-candidate voters a candidate can get is 13. (Please note that non-candidates of a particular department vote as a block).

On similar grounds, we can eliminate Case (iii) as it also implies there is only 1 candidate in FA and MQ.

Now, considering Case (i), we know that 5+0 will happen only one when there are 5 non-candidates in a single department. This is only possible in MS (Out of 7, there will be 2 candidates and 5 non-candidates).

So we can conclude that MS has 2 candidates and that they voted Prof. R.....(i)

We also know that Prof P got 2 votes from Non-candidates. This is only possible when BH has 1 candidate.

So, we can conclude that the number of professors in FA, MS, OQ, BH is 0,2,1,1

	FA	MS	OQ	BH
Total number of voters	9	7	5	3
Number of candidates	0	2	1	1
Number of non-candidates	9	5	4	2

Thus, we get the following table:

	P	Q	R	S
Total Votes	3	14	6	1
Candidate Vote	1(S)	1(R)	1(P)	1(Q)
Votes from non-candidates	2	13	5	0
	BH	FA+OQ	MS	

Now, if we consider Department MS, we know that there are 2 candidates from MS and R can't be one of them as the people in that department voted for him..... (3rd condition).

So the possible combinations of candidates in MS are (P,Q), (Q,S), (P,S).

Now we also know that no one can vote for a candidate in their own department, so we can eliminate (P,S) and (Q,S) as we know that S voted for P and Q voted for S).

So we can infer that P and Q are from MS.

	FA	MS	OQ	BH
Case 1	0	P, Q	R	S
Case 2	0	P, Q	S	R

Now, among the given options, Only Option A is true. Therefore, Option A is the correct answer.

 VIDEO SOLUTION

50. B

Now, we know there is only 1 candidate from OQ, which means that the number of non-candidate voters in OQ will be 4.

We also know that the non-candidates in a particular department voted as a block, and we also know that the least number of non-candidate voters in a particular department can be 1 (BH, 3-2 faculty).

Now, we also know that R got 5 votes from non-candidates.

Now we can write 5 as

i)5

ii)4+1

iii)3+2

Considering case (i) 4+1. This is only possible when there is 1 candidate from OQ, and there are 2 candidates from BH. This implies that the number of candidates in FA and MQ is 1. Now, if we consider FA and MQ and put only 1 candidate there, it implies that there are 15 non-candidate voters between them. Now we know this is not possible since the maximum number of non-candidate voters a candidate can get is 13. (Please note that non-candidates of a particular department vote as a block).

On similar grounds, we can eliminate Case (iii) as it also implies there is only 1 candidate in FA and MQ.

Now, considering Case (i), we know that 5+0 will happen only one when there are 5 non-candidates in a single department. This is only possible in MS (Out of 7, there will be 2 candidates and 5 non-candidates).

So we can conclude that MS has 2 candidates and that they voted Prof. R.....(i)

We also know that Prof P got 2 votes from Non-candidates. This is only possible when BH has 1 candidate.

So, we can conclude that the number of professors in FA, MS, OQ, BH is 0,2,1,1

	FA	MS	OQ	BH
Total number of voters	9	7	5	3
Number of candidates	0	2	1	1
Number of non-candidates	9	5	4	2

Thus, we get the following table:

	P	Q	R	S
Total Votes	3	14	6	1
Candidate Vote	1(S)	1(R)	1(P)	1(Q)
Votes from non candidates	2	13	5	0
	BH	FA+OQ	MS	

Now, if we consider Department MS, we know that there are 2 candidates from MS and R can't be one of them as the people in that department voted for him..... (3rd condition).

So the possible combinations of candidates in MS are (P,Q), (Q,S), (P,S).

Now we also know that no one can vote for a candidate in their own department, so we can eliminate (P,S) and (Q,S) as we know that S voted for P and Q voted for S).

So we can infer that P and Q are from MS.

	FA	MS	OQ	BH
Case 1	0	P, Q	R	S
Case 2	0	P, Q	S	R

From the 2 cases, we can see that the candidate from OQ can either be Prof Ramaswamy or Prof Samuel.

Therefore, the correct answer is Option B.

 VIDEO SOLUTION

100+ CAT DILR Questions



51. B

Now, we know there is only 1 candidate from OQ, which means that the number of non-candidate voters in OQ will be 4.

We also know that the non-candidates in a particular department voted as a block, and we also know that the least number of non-candidate voters in a particular department can be 1 (BH, 3-2 faculty).

Now, we also know that R got 5 votes from non-candidates.

Now we can write 5 as

i) 5

ii) 4+1

iii) 3+2

Considering case (i) 4+1. This is only possible when there is 1 candidate from OQ, and there are 2 candidates from BH. This implies that the number of candidates in FA and MQ is 1. Now, if we consider FA and MQ and put only 1 candidate there, it implies that there are 15 non-candidate voters between them. Now we know this is not possible since the maximum number of non-candidate voters a candidate can get is 13. (Please note that non-candidates of a particular department vote as a block).

On similar grounds, we can eliminate Case (iii) as it also implies there is only 1 candidate in FA and MQ.

Now, considering Case (i), we know that 5+0 will happen only one when there are 5 non-candidates in a single department. This is only possible in MS (Out of 7, there will be 2 candidates and 5 non-candidates).

So we can conclude that MS has 2 candidates and that they voted Prof. R.....(i)

We also know that Prof P got 2 votes from Non-candidates. This is only possible when BH has 1 candidate.

So, we can conclude that the number of professors in FA, MS, OQ, BH is 0,2,1,1

	FA	MS	OQ	BH
Total number of voters	9	7	5	3
Number of candidates	0	2	1	1
Number of non-candidates	9	5	4	2

Thus, we get the following table:

	P	Q	R	S
Total Votes	3	14	6	1
Candidate Vote	1(S)	1(R)	1(P)	1(Q)
Votes from non candidates	2	13	5	0
	BH	FA+OQ	MS	

Now, if we consider Department MS, we know that there are 2 candidates from MS and R can't be one of them as the people in that department voted for him..... (3 rd condition).

So the possible combinations of candidates in MS are (P,Q), (Q,S), (P,S).

Now we also know that no one can vote for a candidate in their own department, so we can eliminate (P,S) and (Q,S) as we know that S voted for P and Q voted for S).

So we can infer that P and Q are from MS.

	FA	MS	OQ	BH
Case 1	0	P, Q	R	S
Case 2	0	P, Q	S	R

Since Prof Qureshi belongs to MS, non-candidates from MS can't vote for him. We can see that the non-candidates from FA voted for him. So, only statement B is true. Therefore, the correct answer is Option B.

 VIDEO SOLUTION

52. D

Now, we know there is only 1 candidate from OQ, which means that the number of non-candidate voters in OQ will be 4.

We also know that the non-candidates in a particular department voted as a block, and we also know that the least number of non-candidate voters in a particular department can be 1 (BH, 3-2 faculty).

Now, we also know that R got 5 votes from non-candidates.

Now we can write 5 as

i)5

ii)4+1

iii)3+2

Considering case (i) 4+1. This is only possible when there is 1 candidate from OQ, and there are 2 candidates from BH. This implies that the number of candidates in FA and MQ is 1. Now, if we consider FA and MQ and put only 1 candidate there, it implies that there are 15 non-candidate voters between them. Now we know this is not possible since the maximum number of non-candidate voters a candidate can get is 13. (Please note that non-candidates of a particular department vote as a block).

On similar grounds, we can eliminate Case (iii) as it also implies there is only 1 candidate in FA and MQ.

Now, considering Case (i), we know that 5+0 will happen only one when there are 5 non-candidates in a single department. This is only possible in MS (Out of 7, there will be 2 candidates and 5 non-candidates).

So we can conclude that MS has 2 candidates and that they voted Prof. R.....(i)

We also know that Prof P got 2 votes from Non-candidates. This is only possible when BH has 1 candidate.

So, we can conclude that the number of professors in FA, MS, OQ, BH is 0,2,1,1

	FA	MS	OQ	BH
Total number of voters	9	7	5	3
Number of candidates	0	2	1	1
Number of non-candidates	9	5	4	2

Thus, we get the following table:

	P	Q	R	S
Total Votes	3	14	6	1
Candidate Vote	1(S)	1(R)	1(P)	1(Q)
Votes from non candidates	2	13	5	0
	BH	FA+OQ	MS	

Now, if we consider Department MS, we know that there are 2 candidates from MS and R can't be one of them as the people in that department voted for him..... (3 rd condition).

So the possible combinations of candidates in MS are (P,Q), (Q,S), (P,S).

Now we also know that no one can vote for a candidate in their own department, so we can eliminate (P,S) and (Q,S) as we know that S voted for P and Q voted for S).

So we can infer that P and Q are from MS.

	FA	MS	OQ	BH
Case 1	0	P, Q	R	S
Case 2	0	P, Q	S	R

We have been told Prof Samuel belongs to B&H. So we have to consider only Case 1. In Case 1 we can see that Prof Prakash belongs to MS and Prof Ramaswamy belongs to OQ.

Therefore, both the statements are true.

So, the correct answer is Option D

[VIDEO SOLUTION](#)

53. B

The bare minimum requirement for the single grandfather and grandmother are as follows:



GF implies Grandfather, GM Grandmother, F Father, M Mother, GS Grandson and GD Granddaughter. The family trees above account for 2 fathers, 2 mothers, 1 GF and 1 GM. Hence, we need 1 more GF, 1 GM, 2 fathers and 2 mothers and two married couples.

The bare minimum tree that would provide this structure is as follows:

GF2 = GM2

F3 = M3

Using these three structures, we have fit all of our requirements. Thus, if GF1, GF2 and GM1 are siblings with the following family trees present, we would have the required number of people in attendance. Hence, the minimum is 12 people.

[VIEW SOLUTION](#)

54. C

A calculator has two memory buttons, A and B. Value 1 is initially stored in both memory locations. The following sequence of steps is carried out five times:

add 1 to B

multiply A to B

store the result in A

What is the value stored in memory location A after this procedure?

The value in A and B before the start of the process is 1 and 1 respectively.

In 1st iteration,

After step I, A = 1 and B = 2

After steps II and III, $A = 1 * 2 = 2$, $B = 2$

In 2nd iteration,

After step I, $A = 2$ and $B = 3$

After steps II and III, $A = 2 * 3 = 6$, $B = 3$

In 3rd iteration,

After step I, $A = 6$ and $B = 4$

After steps II and III, $A = 6 * 4 = 24$, $B = 4$

In 4th iteration,

After step I, $A = 24$ and $B = 5$

After steps II and III, $A = 24 * 5 = 120$, $B = 5$

In 5th iteration,

After step I, $A = 120$ and $B = 6$

After steps II and III, $A = 120 * 6 = 720$, $B = 6$

Hence, option D is the correct answer.

 [VIEW SOLUTION](#)

55. D

From IV: A, B, C, D have one 3, 7, 3, 4-year contract respectively and all other contracts are one-year contracts.

From I, Z has at least one contract every year, the only possible combination is 7+3 or 7+4 year contract and that 7-year contract must be from B.

From III, Vendor W had contracts only in 2012 and from VII, Institutes B and D each had exactly one contract in 2012 $\Rightarrow$ W has got contracts from A and C.

From II, Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that and from VI, VII: C and D didn't have any contract in 2011 and 2010 respectively $\Rightarrow$ A should have X as a 3-year contract from 2010-2012. Now, for 2013-2015 X can't have B for the same. So, X must have got contracts from either C or D in that period.

Case 1:

X has C as a 3-year contract from 2013-2015 but in this case, D can't have any contract in 2012 so, this case is not valid.

Case 2:

X has D for a 4-year contract from 2012-2015 and C must have Z for a three-year contract in the period 2017-2019 such that Z has at least one contract every year.

It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes $\Rightarrow$ A hasn't got any contract from 2013-2019 as it has X, W in the period 2010-2012 and similarly, C shouldn't have any contracts in the years 2010, 2013, 2014, 2015, 2016.

From III, Vendor Y had contracts in 2010 and 2019 and in 2010 D and C hasn't got any contract and A has already got 2 different contracts from two different vendors $\Rightarrow$ Y has a contract from B in 2010 $\Rightarrow$ B hasn't got any contracts in 2017, 2018, 2019.

For Y the only possible contract will be from D $\Rightarrow$ D has got no contracts in the years 2011, 2016, 2017, 2018.

Now, the table looks like:

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
A	X	X	X,W	N	N	N	N	N	N	N
B	Z, Y	Z	Z	Z	Z	Z	Z	N	N	N
C	N	N	W	N	N	N	N	Z	Z	Z
D	N	N	X	X	X	X	N	N	N	Y

'N' represents no contract.

Option D is true.

 VIDEO SOLUTION

100+ CAT VARC Questions



56. C

From IV: A, B, C, D have one 3, 7, 3, 4-year contract respectively and all other contracts are one-year contracts.

From I, Z has at least one contract every year, the only possible combination is 7+3 or 7+4 year contract and that 7-year contract must be from B.

From III, Vendor W had contracts only in 2012 and from VII, Institutes B and D each had exactly one contract in 2012 => W has got contracts from A and C.

From II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that and from VI, VII: C and D didn't have any contract in 2011 and 2010 respectively => A should have X as a 3-year contract from 2010-2012. Now, for 2013-2015 X can't have B for the same. So, X must have got contracts from either C or D in that period.

Case 1:

X has C as a 3-year contract from 2013-2015 but in this case, D can't have any contract in 2012 so, this case is not valid.

Case 2:

X has D for a 4-year contract from 2012-2015 and C must have Z for a three-year contract in the period 2017-2019 such that Z has at least one contract every year.

It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes => A hasn't got any contract from 2013-2019 as it has X, W in the period 2010-2012 and similarly, C shouldn't have any contracts in the years 2010, 2013, 2014, 2015, 2016.

From III, Vendor Y had contracts in 2010 and 2019 and in 2010 D and C hasn't got any contract and A has already got 2 different contracts from two different vendors => Y has a contract from B in 2010 => B hasn't got any contracts in 2017, 2018, 2019.

For Y the only possible contract will be from D => D has got no contracts in the years 2011, 2016, 2017, 2018.

Now, the table looks like:

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
A	X	X	X,W	N	N	N	N	N	N	N
B	Z, Y	Z	Z	Z	Z	Z	Z	N	N	N
C	N	N	W	N	N	N	N	Z	Z	Z
D	N	N	X	X	X	X	N	N	N	Y

'N' represents no contract.

Out of the given options, only 2015 has two contracts and rest have only one contract in that particular year.

 VIDEO SOLUTION

57. A

From IV: A, B, C, D have one 3, 7, 3, 4-year contract respectively and all other contracts are one-year contracts.

From I, Z has at least one contract every year, the only possible combination is 7+3 or 7+4 year contract and that 7-year contract must be from B.

From III, Vendor W had contracts only in 2012 and from VII, Institutes B and D each had exactly one contract in 2012 => W has got contracts from A and C.

From II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that and from VI, VII: C and D didn't have any contract in 2011 and 2010 respectively => A should have X as a 3-year contract from 2010-2012. Now, for 2013-2015 X can't have B for the same. So, X must have got contracts from either C or D in that period.

Case 1:

X has C as a 3-year contract from 2013-2015 but in this case, D can't have any contract in 2012 so, this case is not valid.

Case 2:

X has D for a 4-year contract from 2012-2015 and C must have Z for a three-year contract in the period 2017-2019 such that Z has at least one contract every year.

It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes => A hasn't got any contract from 2013-2019 as it has X, W in the period 2010-2012 and similarly, C shouldn't have any contracts in the years 2010, 2013, 2014, 2015, 2016.

From III, Vendor Y had contracts in 2010 and 2019 and in 2010 D and C hasn't got any contract and A has already got 2 different contracts from two different vendors => Y has a contract from B in 2010 => B hasn't got any contracts in 2017, 2018, 2019.

For Y the only possible contract will be from D => D has got no contracts in the years 2011, 2016, 2017, 2018.

Now, the table looks like:

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
A	X	X	X,W	N	N	N	N	N	N	N
B	Z, Y	Z	Z	Z	Z	Z	Z	N	N	N
C	N	N	W	N	N	N	N	Z	Z	Z
D	N	N	X	X	X	X	N	N	N	Y

'N' represents no contract.

Only during 2016, 2017 and 2018, there was only one contract.

▶ VIDEO SOLUTION

58. B

From IV: A, B, C, D have one 3, 7, 3, 4-year contract respectively and all other contracts are one-year contracts.

From I, Z has at least one contract every year, the only possible combination is 7+3 or 7+4 year contract and that 7-year contract must be from B.

From III, Vendor W had contracts only in 2012 and from VII, Institutes B and D each had exactly one contract in 2012 => W has got contracts from A and C.

From II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that and from VI, VII: C and D didn't have any contract in 2011 and 2010 respectively => A should have X as a 3-year contract from 2010-2012. Now, for 2013-2015 X can't have B for the same. So, X must have got contracts from either C or D in that period.

Case 1:

X has C as a 3-year contract from 2013-2015 but in this case, D can't have any contract in 2012 so, this case is not valid.

Case 2:

X has D for a 4-year contract from 2012-2015 and C must have Z for a three-year contract in the period 2017-2019 such that Z has at least one contract every year.

It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes => A hasn't got any contract from 2013-2019 as it has X, W in the period 2010-2012 and similarly, C shouldn't have any contracts in the years 2010, 2013, 2014, 2015, 2016.

From III, Vendor Y had contracts in 2010 and 2019 and in 2010 D and C hasn't got any contract and A has already got 2 different contracts from two different vendors => Y has a contract from B in 2010 => B hasn't got any contracts in 2017, 2018, 2019.

For Y the only possible contract will be from D => D has got no contracts in the years 2011, 2016, 2017, 2018.

Now, the table looks like:

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
A	X	X	X,W	N	N	N	N	N	N	N
B	Z, Y	Z	Z	Z	Z	Z	Z	N	N	N
C	N	N	W	N	N	N	N	Z	Z	Z
D	N	N	X	X	X	X	N	N	N	Y

'N' represents no contract.

The Number of contracts in 2010 is three.

VIDEO SOLUTION

59. B

From IV: A, B, C, D have one 3, 7, 3, 4-year contract respectively and all other contracts are one-year contracts.

From I, Z has at least one contract every year, the only possible combination is 7+3 or 7+4 year contract and that 7-year contract must be from B.

From III, Vendor W had contracts only in 2012 and from VII, Institutes B and D each had exactly one contract in 2012 => W has got contracts from A and C.

From II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that and from VI, VII: C and D didn't have any contract in 2011 and 2010 respectively => A should have X as a 3-year contract from 2010-2012. Now, for 2013-2015 X can't have B for the same. So, X must have got contracts from either C or D in that period.

Case 1:

X has C as a 3-year contract from 2013-2015 but in this case, D can't have any contract in 2012 so, this case is not valid.

Case 2:

X has D for a 4-year contract from 2012-2015 and C must have Z for a three-year contract in the period 2017-2019 such that Z has at least one contract every year.

It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes => A hasn't got any contract from 2013-2019 as it has X, W in the period 2010-2012 and similarly, C shouldn't have any contracts in the years 2010, 2013, 2014, 2015, 2016.

From III, Vendor Y had contracts in 2010 and 2019 and in 2010 D and C hasn't got any contract and A has already got 2 different contracts from two different vendors => Y has a contract from B in 2010 => B hasn't got any contracts in 2017, 2018, 2019.

For Y the only possible contract will be from D => D has got no contracts in the years 2011, 2016, 2017, 2018.

Now, the table looks like:

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
A	X	X	X,W	N	N	N	N	N	N	N
B	Z, Y	Z	Z	Z	Z	Z	Z	N	N	N
C	N	N	W	N	N	N	N	Z	Z	Z
D	N	N	X	X	X	X	N	N	N	Y

'N' represents no contract.

A, B, W, X had more than one contracts in a single year

 VIDEO SOLUTION

60. C

From IV: A, B, C, D have one 3, 7, 3, 4-year contract respectively and all other contracts are one-year contracts.

From I, Z has at least one contract every year, the only possible combination is 7+3 or 7+4 year contract and that 7-year contract must be from B.

From III, Vendor W had contracts only in 2012 and from VII, Institutes B and D each had exactly one contract in 2012 => W has got contracts from A and C.

From II. Vendor X had one or more contracts in every year up to 2015, but no contract in any year after that and from VI, VII: C and D didn't have any contract in 2011 and 2010 respectively => A should have X as a 3-year contract from 2010-2012. Now, for 2013-2015 X can't have B for the same. So, X must have got contracts from either C or D in that period.

Case 1:

X has C as a 3-year contract from 2013-2015 but in this case, D can't have any contract in 2012 so, this case is not valid.

Case 2:

X has D for a 4-year contract from 2012-2015 and C must have Z for a three-year contract in the period 2017-2019 such that Z has at least one contract every year.

It is known that over the decade, the institutes each got into two contracts with two of these vendors, and each vendor got into two contracts with two of these institutes => A hasn't got any contract from 2013-2019 as it has X, W in the period 2010-2012 and similarly, C shouldn't have any contracts in the years 2010, 2013, 2014, 2015, 2016.

From III, Vendor Y had contracts in 2010 and 2019 and in 2010 D and C hasn't got any contract and A has already got 2 different contracts from two different vendors => Y has a contract from B in 2010 => B hasn't got any contracts in 2017, 2018, 2019.

For Y the only possible contract will be from D => D has got no contracts in the years 2011, 2016, 2017, 2018.

Now, the table looks like:

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
A	X	X	X,W	N	N	N	N	N	N	N
B	Z,Y	Z	Z	Z	Z	Z	Z	N	N	N
C	N	N	W	N	N	N	N	Z	Z	Z
D	N	N	X	X	X	X	N	N	N	Y

'N' represents no contract.

B and A have multiple contracts in a single year.

 VIDEO SOLUTION

CAT Expected Questions PDF



61. A

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	
2014			
2015			
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Bhaskar and Devendra were Gurubhai from 2021 to 2022. Since 2022 is the only option present during this period, it has to be the answer.

Hence, the correct answer is option A.

[VIEW SOLUTION](#)

62. D

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Statement 1) Charu was training under Ustad Samiran in 2018. This is false as she was training under Pandit Meghnath in 2018.

Statement 2) Ananya was training under Ustad Samiran in 2015. This is false as she was training under Pandit Meghnath in 2015.

Statement 3) Ananya was training under Ustad Samiran in 2018. This is false as she was not training in 2018.

Statement 4) Charu was training under Ustad Samiran in 2019. This is true.

Hence, the correct answer is option D.

63.4

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	
2014			
2015			
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Between 2013 and 2024, the years 2017, 2018, 2023, and 2024 were the only years when only two of these four musicians were training under these three Gurus.

Hence, the correct answer is 4.

[VIEW SOLUTION](#)

64. A

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	
2014			
2015			
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can

conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Ananya and Bhaskar were Gurubhai from 2019 to 2020. Since 2020 is the only option present during this period, it has to be the answer.

Hence, the correct answer is option A.

[VIEW SOLUTION](#)

65. B

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	
2014			
2015			
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Charu began her training under Pandit Meghnath in 2015.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)



CAT Important Topics



66.84

In statement 1, it is given that all three of them met the same total number of households, and each of them sold a total of 100 items in two days. In statement 2, it is given that on both days, Lahur met the same number of households and sold the same number of items. This implies he sold 50 items per day. Let the number households Lahur met in a day be 'x'.

Total number of households each of them met in two days will be '2x'.

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri					2x	100
Hokli					2x	100
Lahur	x	50	x	50	2x	100

In statement 3, it is given that Hokli could not sell any item on the second day because the first household he met on that day complained against him. This implies he met only 1 household on day 2.

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri					2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 4, it is given that Tohri met 30 more households on the second day than on the first day.

Let the number of households Tohri met on day 1 be 'a'

It is given,

$$a + a + 30 = 2x$$

$$a + 15 = x$$

$$a = x - 15$$

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri	x-15	y	x+15	100-y	2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 5, it is given that

$$2 \left(\frac{50}{x} \right) = \frac{y}{x-15} \dots\dots (1)$$

$$\frac{3}{4} \left(\frac{50}{x} \right) = \frac{100-y}{x+15} \dots\dots (2)$$

$$\frac{100}{x} = \frac{y}{x-15}$$

$$\frac{100}{y} = \frac{x}{x-15}$$

$$\frac{y}{100} = 1 - \frac{15}{x}$$

$$x = \frac{1500}{100-y}$$

Substituting x in (2), we get

$$y = 40 \text{ and } x = 25$$

Final Table:

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri	10	40	40	60	50	100
Hokli	49	100	1	0	50	100
Lahur	25	50	25	50	50	100

The total number of households met by Tohri, Hokli and Lahur on the first day is 10 + 49 + 25, i.e. 84.

 VIDEO SOLUTION

67. D

In statement 1, it is given that all three of them met the same total number of households, and each of them sold a total of 100 items in two days. In statement 2, it is given that on both days, Lahur met the same number of households and sold the same number of items. This implies he sold 50 items per day. Let the number households Lahur met in a day be 'x'.

Total number of households each of them met in two days will be '2x'.

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri					2x	100
Hokli					2x	100
Lahur	x	50	x	50	2x	100

In statement 3, it is given that Hokli could not sell any item on the second day because the first household he met on that day complained against him. This implies he met only 1 household on day 2.

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri					2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 4, it is given that Tohri met 30 more households on the second day than on the first day.

Let the number of households Tohri met on day 1 be 'a'

It is given,

$$a + a + 30 = 2x$$

$$a + 15 = x$$

$$a = x - 15$$

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri	x-15	y	x+15	100-y	2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 5, it is given that

$$2 \left(\frac{50}{x} \right) = \frac{y}{x-15} \dots\dots (1)$$

$$\frac{3}{4} \left(\frac{50}{x} \right) = \frac{100-y}{x+15} \dots\dots (2)$$

$$\frac{100}{x} = \frac{y}{x-15}$$

$$\frac{100}{y} = \frac{x}{x-15}$$

$$\frac{y}{100} = 1 - \frac{15}{x}$$

$$x = \frac{1500}{100-y}$$

Substituting x in (2), we get

$$y = 40 \text{ and } x = 25$$

Final Table:

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri	10	40	40	60	50	100
Hokli	49	100	1	0	50	100
Lahur	25	50	25	50	50	100

Tohri met 10 households on day 1. The answer is option D.

 VIDEO SOLUTION

68. A

In statement 1, it is given that all three of them met the same total number of households, and each of them sold a total of 100 items in two days. In statement 2, it is given that on both days, Lahur met the same number of households and sold the same number of items. This implies he sold 50 items per day. Let the number households Lahur met in a day be 'x'.

Total number of households each of them met in two days will be '2x'.

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri					2x	100
Hokli					2x	100
Lahur	x	50	x	50	2x	100

In statement 3, it is given that Hokli could not sell any item on the second day because the first household he met on that day complained against him. This implies he met only 1 household on day 2.

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri					2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 4, it is given that Tohri met 30 more households on the second day than on the first day.

Let the number of households Tohri met on day 1 be 'a'

It is given,

$$a + a + 30 = 2x$$

$$a + 15 = x$$

$$a = x - 15$$

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri	x-15	y	x+15	100-y	2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 5, it is given that

$$2 \left(\frac{50}{x} \right) = \frac{y}{x-15} \dots\dots (1)$$

$$\frac{3}{4} \left(\frac{50}{x} \right) = \frac{100-y}{x+15} \dots\dots (2)$$

$$\frac{100}{x} = \frac{y}{x-15}$$

$$\frac{100}{y} = \frac{x}{x-15}$$

$$\frac{y}{100} = 1 - \frac{15}{x}$$

$$x = \frac{1500}{100-y}$$

Substituting x in (2), we get

$$y = 40 \text{ and } x = 25$$

Final Table:

	Day 1		Day 2		Total	
	HHS	Items	HHS	Items	HHS	Items
Tohri	10	40	40	60	50	100
Hokli	49	100	1	0	50	100
Lahur	25	50	25	50	50	100

Among the three, Tohri had the highest success rate on the second day - this statement is incorrect. On day 2, Lahur had the highest success rate, i.e. 2 whereas Tohri's success rate is 1.5.

Tohri had a higher success rate on the first day compared to the second day - this statement is correct.

Tohri's day 1 success rate is 4 and day 2 success rate is 1.5.

Among the three, Tohri had the highest success rate on the first day - this statement is correct.

Tohri's success rate on day 1 is 4.

Hokli's success rate on day 1 is 2.04.

Lahur's success rate on day 1 is 2.

Among the three, Lahur had the lowest success rate on the first day - this statement is correct.

Tohri's success rate on day 1 is 4.

Hokli's success rate on day 1 is 2.04.

Lahur's success rate on day 1 is 2.

The answer is option A.

69. A

In statement 1, it is given that all three of them met the same total number of households, and each of them sold a total of 100 items in two days. In statement 2, it is given that on both days, Lahur met the same number of households and sold the same number of items. This implies he sold 50 items per day. Let the number of households Lahur met in a day be 'x'.

Total number of households each of them met in two days will be '2x'.

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri					2x	100
Hokli					2x	100
Lahur	x	50	x	50	2x	100

In statement 3, it is given that Hokli could not sell any item on the second day because the first household he met on that day complained against him. This implies he met only 1 household on day 2.

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri					2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 4, it is given that Tohri met 30 more households on the second day than on the first day.

Let the number of households Tohri met on day 1 be 'a'

It is given,

$$a + a + 30 = 2x$$

$$a + 15 = x$$

$$a = x - 15$$

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri	x-15	y	x+15	100-y	2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 5, it is given that

$$2 \left(\frac{50}{x} \right) = \frac{y}{x-15} \dots\dots (1)$$

$$\frac{3}{4} \left(\frac{50}{x} \right) = \frac{100-y}{x+15} \dots\dots (2)$$

$$\frac{100}{x} = \frac{y}{x-15}$$

$$\frac{100}{y} = \frac{x}{x-15}$$

$$\frac{y}{100} = 1 - \frac{15}{x}$$

$$x = \frac{1500}{100-y}$$

Substituting x in (2), we get

$$y = 40 \text{ and } x = 25$$

Final Table:

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri	10	40	40	60	50	100
Hokli	49	100	1	0	50	100
Lahur	25	50	25	50	50	100

Lahur met 25 households on day 2. The answer is option A.

70.40

In statement 1, it is given that all three of them met the same total number of households, and each of them sold a total of 100 items in two days. In statement 2, it is given that on both days, Lahur met the same number of households and sold the same number of items. This implies he sold 50 items per day. Let the number of households Lahur met in a day be 'x'.

Total number of households each of them met in two days will be '2x'.

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri					2x	100
Hokli					2x	100
Lahur	x	50	x	50	2x	100

In statement 3, it is given that Hokli could not sell any item on the second day because the first household he met on that day complained against him. This implies he met only 1 household on day 2.

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri					2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 4, it is given that Tohri met 30 more households on the second day than on the first day.

Let the number of households Tohri met on day 1 be 'a'

It is given,

$$a + a + 30 = 2x$$

$$a + 15 = x$$

$$a = x - 15$$

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri	x-15	y	x+15	100-y	2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 5, it is given that

$$2 \left(\frac{50}{x} \right) = \frac{y}{x-15} \dots\dots (1)$$

$$\frac{3}{4} \left(\frac{50}{x} \right) = \frac{100-y}{x+15} \dots\dots (2)$$

$$\frac{100}{x} = \frac{y}{x-15}$$

$$\frac{100}{y} = \frac{x}{x-15}$$

$$\frac{y}{100} = 1 - \frac{15}{x}$$

$$x = \frac{1500}{100-y}$$

Substituting x in (2), we get

$$y = 40 \text{ and } x = 25$$

Final Table:

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri	10	40	40	60	50	100
Hokli	49	100	1	0	50	100
Lahur	25	50	25	50	50	100

The number of items sold by Tohri on the first day is 40.





71.2

We are given that the numbers keep increasing from left to right (clue 1), and the number keeps decreasing from top to bottom (clue 2)

The key takeaway from this is that 10 must be placed in Row 1, column 4, as placing it anywhere else would mean that the number above it, or right to it, must be greater than 10, which is not an option.

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2				
Row 3				
Row 4				

Clue 3 says that one is either in the same row or the same column as 10.

The same logic that we used for 10 applies for 1; it must be either in Row 1, column 1, or Row 4, column 4; we don't know which one yet.

We are given that 2 and 3 are not in the same column or row as 10, meaning that they must occupy two spots from (Row 2, Col. 2), (Row 2, Col. 3), and (Row 3, Col. 3)

2 and 3 must be present in Row 2, column 2 or Row 3, column 3, as there is no number smaller than it to be in the cell left to it.

Clue 6 says that 4 and 6 are in the same row; this can be rows 1, 2, or 3.

Clue 5 is a good starting point.

Once we found the position of 10, the only positions possible for 9 are R1C3 or R2C4

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2		2/3		9
Row 3			2/3	
Row 4				

If 9 is placed in R2C4, 7 and 8 must be placed in row 1.

Clue 6 says that 4 and 6 are in the same row, but with 7 and 8 in row 1, no rows are left with two spaces. Hence, 9 can not be in R2C4, and the arrangement must be:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

7 and 8 must be in column 4, occupying R3C4 and R2C4, respectively.

Hence, 4, 6 must be in row 1:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1	4	6	9	10
Row 2		2/3	5	8
Row 3			2/3	7
Row 4				1

Placements of all the numbers except 2 and 3 can be determined.

Therefore, 2 is the correct answer.

▶ VIDEO SOLUTION

72.26

We are given that the numbers keep increasing from left to right (clue 1), and the number keeps decreasing from top to bottom (clue 2)

The key takeaway from this is that 10 must be placed in Row 1, column 4, as placing it anywhere else would mean that the number above it, or right to it, must be greater than 10, which is not an option.

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2				
Row 3				
Row 4				

Clue 3 says that one is either in the same row or the same column as 10.

The same logic that we used for 10 applies for 1; it must be either in Row 1, column 1, or Row 4, column 4; we don't know which one yet.

We are given that 2 and 3 are not in the same column or row as 10, meaning that they must occupy two spots from (Row 2, Col. 2), (Row 2, Col. 3), and (Row 3, Col. 3)

2 and 3 must be present in Row 2, column 2 or Row 3, column 3, as there is no number smaller than it to be in the cell left to it.

Clue 6 says that 4 and 6 are in the same row; this can be rows 1, 2, or 3.

Clue 5 is a good starting point.

Once we found the position of 10, the only positions possible for 9 are R1C3 or R2C4

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2		2/3		9
Row 3			2/3	
Row 4				

If 9 is placed in R2C4, 7 and 8 must be placed in row 1.

Clue 6 says that 4 and 6 are in the same row, but with 7 and 8 in row 1, no rows are left with two spaces. Hence, 9 can not be in R2C4, and the arrangement must be:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

7 and 8 must be in column 4, occupying R3C4 and R2C4, respectively.

Hence, 4, 6 must be in row 1:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1	4	6	9	10
Row 2		2/3	5	8
Row 3			2/3	7
Row 4				1

The sum of numbers in the 4th column is $10+8+7+1 = 26$

Therefore, 26 is the correct answer.

73.C

We are given that the numbers keep increasing from left to right (clue 1), and the number keeps decreasing from top to bottom (clue 2)

The key takeaway from this is that 10 must be placed in Row 1, column 4, as placing it anywhere else would mean that the number above it, or right to it, must be greater than 10, which is not an option.

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2				
Row 3				
Row 4				

Clue 3 says that one is either in the same row or the same column as 10.

The same logic that we used for 10 applies for 1; it must be either in Row 1, column 1, or Row 4, column 4; we don't know which one yet.

We are given that 2 and 3 are not in the same column or row as 10, meaning that they must occupy two spots from (Row 2, Col. 2), (Row 2, Col. 3), and (Row 3, Col. 3)

2 and 3 must be present in Row 2, column 2 or Row 3, column 3, as there is no number smaller than it to be in the cell left to it.

Clue 6 says that 4 and 6 are in the same row; this can be rows 1, 2, or 3.

Clue 5 is a good starting point.

Once we found the position of 10, the only positions possible for 9 are R1C3 or R2C4

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2		2/3		9
Row 3			2/3	
Row 4				

If 9 is placed in R2C4, 7 and 8 must be placed in row 1.

Clue 6 says that 4 and 6 are in the same row, but with 7 and 8 in row 1, no rows are left with two spaces. Hence, 9 can not be in R2C4, and the arrangement must be:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

7 and 8 must be in column 4, occupying R3C4 and R2C4, respectively.

Hence, 4, 6 must be in row 1:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1	4	6	9	10
Row 2		2/3	5	8
Row 3			2/3	7
Row 4				1

Positions 2 and 3 can be switched.

Either of them can be in column 2 or 3

Neither of the statements is necessarily true.

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

74.4

We are given that the numbers keep increasing from left to right (clue 1), and the number keeps decreasing from top to bottom (clue 2)

The key takeaway from this is that 10 must be placed in Row 1, column 4, as placing it anywhere else would mean that the number above it, or right to it, must be greater than 10, which is not an option.

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2				
Row 3				
Row 4				

Clue 3 says that one is either in the same row or the same column as 10.

The same logic that we used for 10 applies for 1; it must be either in Row 1, column 1, or Row 4, column 4; we don't know which one yet.

We are given that 2 and 3 are not in the same column or row as 10, meaning that they must occupy two spots from (Row 2, Col. 2), (Row 2, Col. 3), and (Row 3, Col. 3)

2 and 3 must be present in Row 2, column 2 or Row 3, column 3, as there is no number smaller than it to be in the cell left to it.

Clue 6 says that 4 and 6 are in the same row; this can be rows 1, 2, or 3.

Clue 5 is a good starting point.

Once we found the position of 10, the only positions possible for 9 are R1C3 or R2C4

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2		2/3		9
Row 3			2/3	
Row 4				

If 9 is placed in R2C4, 7 and 8 must be placed in row 1.

Clue 6 says that 4 and 6 are in the same row, but with 7 and 8 in row 1, no rows are left with two spaces. Hence, 9 can not be in R2C4, and the arrangement must be:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

7 and 8 must be in column 4, occupying R3C4 and R2C4, respectively.

Hence, 4, 6 must be in row 1:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1	4	6	9	10
Row 2		2/3	5	8
Row 3			2/3	7
Row 4				1

The lowest sum is of the 4th row.

Therefore, is the correct answer.

VIDEO SOLUTION

75. A

We are given that the numbers keep increasing from left to right (clue 1), and the number keeps decreasing from top to bottom (clue 2)

The key takeaway from this is that 10 must be placed in Row 1, column 4, as placing it anywhere else would mean that the number above it, or right to it, must be greater than 10, which is not an option.

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2				
Row 3				
Row 4				

Clue 3 says that one is either in the same row or the same column as 10.

The same logic that we used for 10 applies for 1; it must be either in Row 1, column 1, or Row 4, column 4; we don't know which one yet.

We are given that 2 and 3 are not in the same column or row as 10, meaning that they must occupy two spots from (Row 2, Col. 2), (Row 2, Col. 3), and (Row 3, Col. 3)

2 and 3 must be present in Row 2, column 2 or Row 3, column 3, as there is no number smaller than it to be in the cell left to it.

Clue 6 says that 4 and 6 are in the same row; this can be rows 1, 2, or 3.

Clue 5 is a good starting point.

Once we found the position of 10, the only positions possible for 9 are R1C3 or R2C4

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1				10
Row 2		2/3		9
Row 3			2/3	
Row 4				

If 9 is placed in R2C4, 7 and 8 must be placed in row 1.

Clue 6 says that 4 and 6 are in the same row, but with 7 and 8 in row 1, no rows are left with two spaces. Hence, 9 can not be in R2C4, and the arrangement must be:

	Col. 1	Col. 2	Col. 3	Col. 4
Row 1			9	10
Row 2		2/3		
Row 3			2/3	
Row 4				

7 and 8 must be in column 4, occupying R3C4 and R2C4, respectively.

Hence, 4, 6 must be in row 1:

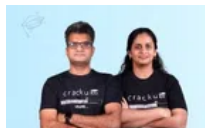
	Col. 1	Col. 2	Col. 3	Col. 4
Row 1	4	6	9	10
Row 2		2/3	5	8
Row 3			2/3	7
Row 4				1

We can see that the positions 10 and 1 are fixed and agree with the given statements.

Both the statements are true.

Therefore, Option A is the correct answer.

[VIDEO SOLUTION](#)



CAT Mock Test

Mocks designed exactly like the actual CAT




76.6

We are told that each coach had at least two players. Clue 1 says that Xena trained more people than Yuki.

It must be the case that Yuki trained only two people. If Yuki had trained three, then Xena would have trained at least four players, leaving only one for Zara.

Hence, Yuki trained two people.

Coming to the scores given to the players themselves:

We are given that only 5 and 7 received a same rating; everyone else received a distinct rating.

We are also given their average to be 4 (clue 4), giving the sum of all the scores they got to be 32

The sum of all numbers from 1 to 7 would be $\frac{8}{2} \times 7 = 28$, hence the repeated score must be $32 - 28 = 4$

Thus, the score of 5 and 7 was 4

Clue 5 says that player 2 got the highest score, which is 7

Clue 7 says that player 4 got a score double that of player 8 but less than player 5; the only possibility is player 4 getting 2 and player 8 getting 1.

Player	Score
1	
2	7
3	
4	2
5	4
6	
7	4
8	1

Now, considering clues 2, 3 and 6:

We are given the same coach trained in 5 and 7. And 2, 3, and 5 were trained by different coaches, with 1 and 4 being trained by the same coach.

Cheer 6 informs us about the average number of players the coaches train.

We know that Yuki trained only two players.

Let's take the average score of Yuki's players to be x

The average of Xena's players would be $x/2$, and that of Zara's players would be $x-2$

The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $3x/2$ or $2x$.

The total score of Zara's players would be $3x-6$ or $2x-4$

We have two cases to consider:



Let's say Xena and Zara had three players each.

The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $3x/2$.

The total score of Zara's players would be $3x-6$.

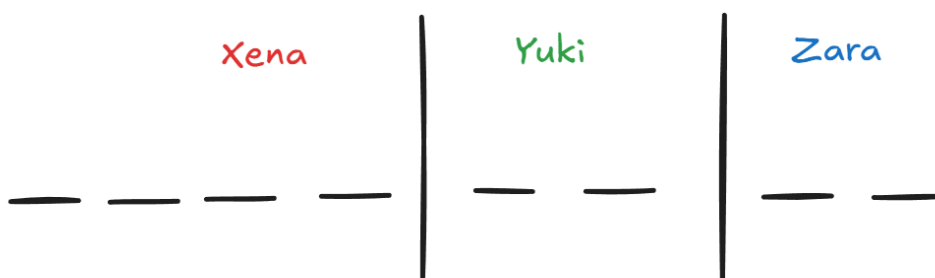
The sum of all these scores would be $\frac{13x}{2} - 6$ which should be equal to 32

This would give us the value of x as $\frac{13x}{2} = 38$

Which would give a non-integral value of $2x$, that is, the sum of Yuki's player's score.

Hence, this must not be the case.

The other possibility:



The total score of Yuki's players would be $2x$.
 The total score of Xena's players would be $2x$.
 The total score of Zara's players would be $2x-4$.

The sum of all these scores would be $6x-4$, which should be equal to 32
 This would give the value of x as 6

Hence, the sum of all of Yuki's players would be 12
 The sum of all of Xena's players would be 12
 The sum of all of Zara's players would be 8

Yuki has only two players whose scores add up to 12; the only combination possible is scores 7+5, where seven were scored by player 2. Hence, player two must be under Yuki.

Zara got a total of 8 scores, with 7 and 5 gone. The combinations that could get this score are 2+6 and 4+4
 The score of 2 is obtained by player 4, which must come with player 1

It is possible that 1 could have gotten a score 6

But then we run into a contradiction: players 3 and 5 would end up under the same coach.

Hence, Zara must have gotten 8 through 4+4 with players 5 and 7.

All the remaining scores must be with Xena, adding up to 12, which is the case.

4 and 1 must be present together, and 3 must be present in Xena as well, giving us the arrangement.

	Xena				Yuki		Zara	
Players	8	4	1/3	1/3	2	6	5	7
Score:	<u>1</u>	<u>2</u>	<u>3</u>	<u>6</u>	<u>7</u>	<u>5</u>	<u>4</u>	<u>4</u>

We can determine the ratings of all the players except 1 and 3

Therefore, 6 is the correct answer.

[VIDEO SOLUTION](#)

77.4

We are told that each coach had at least two players. Clue 1 says that Xena trained more people than Yuki. It must be the case that Yuki trained only two people. If Yuki had trained three, then Xena would have trained at least four players, leaving only one for Zara.

Hence, Yuki trained two people.

Coming to the scores given to the players themselves:

We are given that only 5 and 7 received a sam rating; everyone else received a distinct rating.

We are also given their average to be 4 (clue 4), giving the sum of all the scores they got to be 32

The sum of all numbers from 1 to 7 would be $\frac{8}{2} \times 7 = 28$, hence the repeated score must be $32-28 = 4$

Thus, the score of 5 and 7 was 4

Clue 5 says that player 2 got the highest score, which is 7

Clue 7 says that player 4 got a score double that of player 8 but less than player 5; the only possibility is player 4 getting 2 and player 8 getting 1.

Player	Score
1	
2	7
3	
4	2
5	4
6	
7	4
8	1

Now, considering clues 2, 3 and 6:

We are given the same coach trained in 5 and 7. And 2, 3, and 5 were trained by different coaches, with 1 and 4 being trained by the same coach.

Cheer 6 informs us about the average number of players the coaches train.

We know that Yuki trained only two players.

Let's take the average score of Yuki's players to be x

The average of Xena's players would be $x/2$, and that of Zara's players would be $x-2$

The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $3x/2$ or $2x$.

The total score of Zara's players would be $3x-6$ or $2x-4$

We have two cases to consider:



Let's say Xena and Zara had three players each.

The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $3x/2$.

The total score of Zara's players would be $3x-6$.

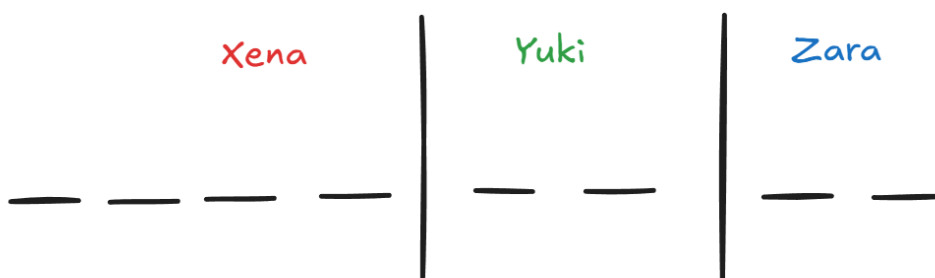
The sum of all these scores would be $\frac{13x}{2} - 6$ which should be equal to 32

This would give us the value of x as $\frac{13x}{2} = 38$

Which would give a non-integral value of $2x$, that is, the sum of Yuki's player's score.

Hence, this must not be the case.

The other possibility:



The total score of Yuki's players would be $2x$.
 The total score of Xena's players would be $2x$.
 The total score of Zara's players would be $2x-4$.

The sum of all these scores would be $6x-4$, which should be equal to 32
 This would give the value of x as 6

Hence, the sum of all of Yuki's players would be 12
 The sum of all of Xena's players would be 12
 The sum of all of Zara's players would be 8

Yuki has only two players whose scores add up to 12; the only combination possible is scores 7+5, where seven were scored by player 2. Hence, player two must be under Yuki.

Zara got a total of 8 scores, with 7 and 5 gone. The combinations that could get this score are 2+6 and 4+4
 The score of 2 is obtained by player 4, which must come with player 1

It is possible that 1 could have gotten a score 6

But then we run into a contradiction: players 3 and 5 would end up under the same coach.

Hence, Zara must have gotten 8 through 4+4 with players 5 and 7.

All the remaining scores must be with Xena, adding up to 12, which is the case.

4 and 1 must be present together, and 3 must be present in Xena as well, giving us the arrangement.

	Xena				Yuki		Zara	
Players	8	4	1/3	1/3	2	6	5	7
Score:	<u>1</u>	<u>2</u>	<u>3</u>	<u>6</u>	<u>7</u>	<u>5</u>	<u>4</u>	<u>4</u>

Score of player 7 was 4

Therefore, 4 is the correct answer.

[VIDEO SOLUTION](#)

78. B

We are told that each coach had at least two players. Clue 1 says that Xena trained more people than Yuki. It must be the case that Yuki trained only two people. If Yuki had trained three, then Xena would have trained at least four players, leaving only one for Zara.

Hence, Yuki trained two people.

Coming to the scores given to the players themselves:

We are given that only 5 and 7 received a same rating; everyone else received a distinct rating.

We are also given their average to be 4 (clue 4), giving the sum of all the scores they got to be 32

The sum of all numbers from 1 to 7 would be $\frac{8}{2} \times 7 = 28$, hence the repeated score must be $32-28 = 4$

Thus, the score of 5 and 7 was 4

Clue 5 says that player 2 got the highest score, which is 7

Clue 7 says that player 4 got a score double that of player 8 but less than player 5; the only possibility is player 4 getting 2 and player 8 getting 1.

Player	Score
1	
2	7
3	
4	2
5	4
6	
7	4
8	1

Now, considering clues 2, 3 and 6:

We are given the same coach trained in 5 and 7. And 2, 3, and 5 were trained by different coaches, with 1 and 4 being trained by the same coach.

Cheer 6 informs us about the average number of players the coaches train.

We know that Yuki trained only two players.

Let's take the average score of Yuki's players to be x

The average of Xena's players would be $x/2$, and that of Zara's players would be $x-2$

The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $3x/2$ or $2x$.

The total score of Zara's players would be $3x-6$ or $2x-4$

We have two cases to consider:



Let's say Xena and Zara had three players each.

The total score of Yuki's players would be $2x$.

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The total score of Zara's players would be $3x-6$.

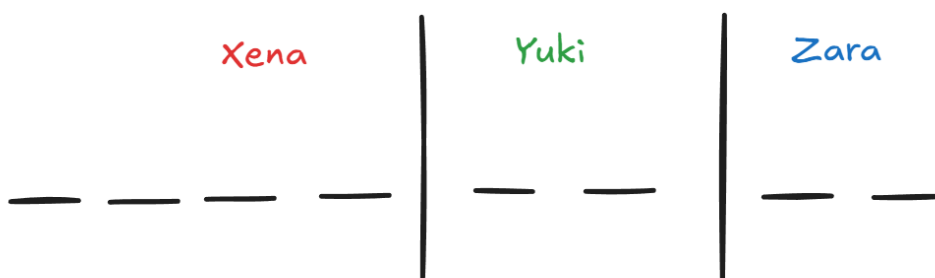
The sum of all these scores would be $\frac{13x}{2} - 6$ which should be equal to 32

This would give us the value of x as $\frac{13x}{2} = 38$

Which would give a non-integral value of $2x$, that is, the sum of Yuki's player's score.

Hence, this must not be the case.

The other possibility:



The total score of Yuki's players would be $2x$.
 The total score of Xena's players would be $2x$.
 The total score of Zara's players would be $2x-4$.

The sum of all these scores would be $6x-4$, which should be equal to 32
 This would give the value of x as 6

Hence, the sum of all of Yuki's players would be 12
 The sum of all of Xena's players would be 12
 The sum of all of Zara's players would be 8

Yuki has only two players whose scores add up to 12; the only combination possible is scores 7+5, where seven were scored by player 2. Hence, player two must be under Yuki.

Zara got a total of 8 scores, with 7 and 5 gone. The combinations that could get this score are 2+6 and 4+4
 The score of 2 is obtained by player 4, which must come with player 1

It is possible that 1 could have gotten a score 6

But then we run into a contradiction: players 3 and 5 would end up under the same coach.

Hence, Zara must have gotten 8 through 4+4 with players 5 and 7.

All the remaining scores must be with Xena, adding up to 12, which is the case.

4 and 1 must be present together, and 3 must be present in Xena as well, giving us the arrangement.

	Xena				Yuki		Zara	
Players	8	4	1/3	1/3	2	6	5	7
Score:	<u>1</u>	<u>2</u>	<u>3</u>	<u>6</u>	<u>7</u>	<u>5</u>	<u>4</u>	<u>4</u>

We can see that Zara had two players.

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

79.5

We are told that each coach had at least two players. Clue 1 says that Xena trained more people than Yuki. It must be the case that Yuki trained only two people. If Yuki had trained three, then Xena would have trained at least four players, leaving only one for Zara.

Hence, Yuki trained two people.

Coming to the scores given to the players themselves:

We are given that only 5 and 7 received a same rating; everyone else received a distinct rating.

We are also given their average to be 4 (clue 4), giving the sum of all the scores they got to be 32

The sum of all numbers from 1 to 7 would be $\frac{8}{2} \times 7 = 28$, hence the repeated score must be $32-28 = 4$

Thus, the score of 5 and 7 was 4

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Clue 7 says that player 4 got a score double that of player 8 but less than player 5; the only possibility is player 4 getting 2 and player 8 getting 1.

Player	Score
1	
2	7
3	
4	2
5	4
6	
7	4
8	1

Now, considering clues 2, 3 and 6:

We are given the same coach trained in 5 and 7. And 2, 3, and 5 were trained by different coaches, with 1 and 4 being trained by the same coach.

Cheer 6 informs us about the average number of players the coaches train.

We know that Yuki trained only two players.

Let's take the average score of Yuki's players to be x

The average of Xena's players would be $x/2$, and that of Zara's players would be $x-2$

The total score of Yuki's players would be $2x$.

The total score of Xena's players would be $3x/2$ or $2x$.

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We have two cases to consider:



Let's say Xena and Zara had three players each.

The total score of Yuki's players would be $2x$.

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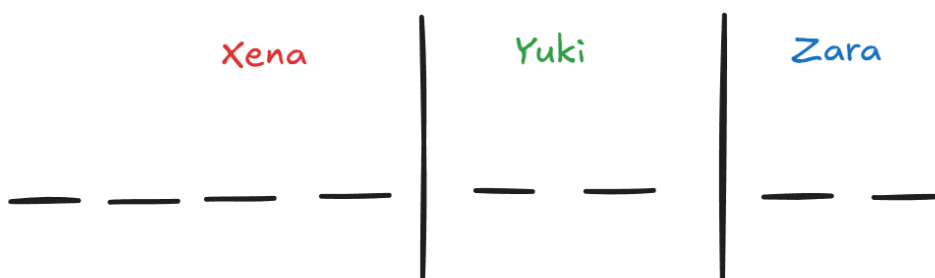
The sum of all these scores would be $\frac{13x}{2} - 6$ which should be equal to 32

This would give us the value of x as $\frac{13x}{2} = 38$

Which would give a non-integral value of $2x$, that is, the sum of Yuki's player's score.

Hence, this must not be the case.

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The total score of Yuki's players would be $2x$.
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The sum of all these scores would be $6x-4$, which should be equal to 32
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Hence, the sum of all of Yuki's players would be 12
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Yuki has only two players whose scores add up to 12; the only combination possible is scores 7+5, where seven were scored by player 2. Hence, player two must be under Yuki.

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 The score of 2 is obtained by player 4, which must come with player 1

It is possible that 1 could have gotten a score 6

But then we run into a contradiction: players 3 and 5 would end up under the same coach.

Hence, Zara must have gotten 8 through 4+4 with players 5 and 7.

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	Xena				Yuki		Zara	
Players	8	4	1/3	1/3	2	6	5	7
Score:	<u>1</u>	<u>2</u>	<u>3</u>	<u>6</u>	<u>7</u>	<u>5</u>	<u>4</u>	<u>4</u>

The rating of player 6 was 5

Therefore, 5 is the correct answer.

[VIDEO SOLUTION](#)

80. B

We are told that each coach had at least two players. Clue 1 says that Xena trained more people than Yuki. It must be the case that Yuki trained only two people. If Yuki had trained three, then Xena would have trained at least four players, leaving only one for Zara.

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1	
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3	
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6	
7	4
8	1

Now, considering clues 2, 3 and 6:

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The total score of Yuki's players would be $2x$.

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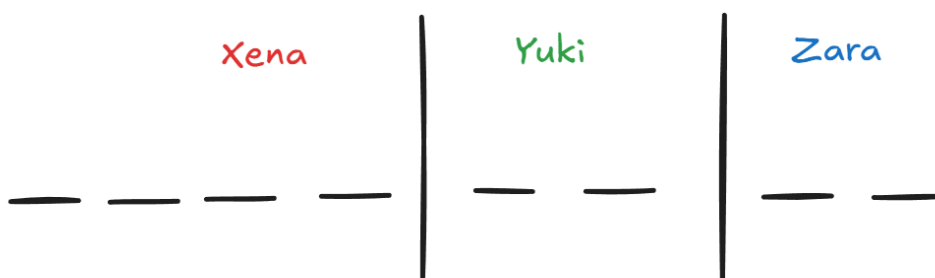
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 The total score of Zara's players would be $2x-4$.

The sum of all these scores would be $6x-4$, which should be equal to 32
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 The sum of all of Xena's players would be 12
 The sum of all of Zara's players would be 8

Yuki has only two players whose scores add up to 12; the only combination possible is scores 7+5, where seven were scored by player 2. Hence, player two must be under Yuki.

Zara got a total of 8 scores, with 7 and 5 gone. The combinations that could get this score are 2+6 and 4+4
 The score of 2 is obtained by player 4, which must come with player 1

It is possible that 1 could have gotten a score 6

But then we run into a contradiction: players 3 and 5 would end up under the same coach.

Hence, Zara must have gotten 8 through 4+4 with players 5 and 7.

All the remaining scores must be with Xena, adding up to 12, which is the case.

4 and 1 must be present together, and 3 must be present in Xena as well, giving us the arrangement.

	Xena				Yuki		Zara	
Players	8	4	1/3	1/3	2	6	5	7
Score:	<u>1</u>	<u>2</u>	<u>3</u>	<u>6</u>	<u>7</u>	<u>5</u>	<u>4</u>	<u>4</u>

Xena coached players numbered 1, 3, 4 and 8

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)



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